Stochastic Modeling and Simulation of Vehicle Headways on Highways: A Bayesian Headway Distribution, Renewal Arrival Process, and Regime Markov Chain Using highD and NGSIM Trajectory Datasets

Sanidhya Joshi (Team Leader) Roll No. S20240010214
Kaustabh Kadam Roll No. S20240010114
Dev Dharsandiya Roll No. S20240010058
International Institute of Information Technology, Sri City, India

Abstract

The distribution assumed for vehicle time headways determines how highway capacity, safety margins and the variability of arrival counts are modelled, yet no single standard distribution fits all traffic conditions. This paper studies headways as a stochastic system on two public trajectory datasets: highD (1,225,460 one-second headways, German highways) and NGSIM I-80 (117,068, California). A recently proposed two-parameter density, $f(t \mid a, b) \propto b^{|t-a|}$, is compared with six standard distributions by Bayesian estimation with a Metropolis–Hastings sampler; arrivals are modelled as a headway-driven renewal counting process, and car-following regimes as a three-state Markov chain estimated from within-trajectory transitions only. With the published likelihood, which ignores the truncation of the data to $[0.5, 25]$ s, the proposed distribution has the lowest Kullback–Leibler divergence on highD (0.0036) and ranks fourth on NGSIM (0.0760; Burr is best, 0.0015), as published. Conditioning the likelihood on the observation window changes the highD ranking: the shifted log-normal becomes best (0.0026) and the proposed distribution second. All 28 posteriors converge (split $\hat{R} \leq 1.006$), including the highD shifted exponential, whose earlier non-convergence ($\hat{R} = 4.92$) is removed by reparametrising its threshold. Simulated renewal counts agree with renewal theory (mean error at most 0.2%, variance error at most 7.0%), and the headway-driven model implies arrivals more regular than Poisson (variance-to-mean ratios 0.61 and 0.46). Detector inter-arrival times reconstructed from the trajectories qualify this: on highD, one-minute counts have a variance-to-mean ratio of 0.96, close to Poisson, and the within-trajectory headway sequence is strongly autocorrelated (lag-1 coefficient 0.906). The study separates mathematical, simulation and empirical validation and documents where published conclusions depend on estimation choices.

Keywords: Traffic Headway; Stochastic Modelling; Bayesian Inference; Metropolis–Hastings Algorithm; Renewal Process; Markov Chain; Traffic Flow; highD Dataset; NGSIM Dataset.

1 Introduction

1.1 Background and motivation

The time headway between a vehicle and the vehicle ahead of it in the same lane is one of the basic microscopic variables of traffic flow theory. The reciprocal of its mean is the flow that a lane can carry in car-following conditions, short headways reduce the margin for braking, and

its distribution is an input to capacity analysis, signal timing and microscopic simulation [1–3]. Headways are random. They depend on driver behaviour, vehicle type, speed, traffic volume and road geometry, so that no single deterministic value describes a lane, and the shape of the distribution changes between free-flowing and congested traffic. A large literature has therefore proposed and compared headway distributions without converging on a single model [1, 4].

Leungbootnak et al. [4] proposed a new two-parameter headway density, $f(t \mid a, b) \propto b^{|t-a|}$, and compared it with six standard distributions on five trajectory datasets. This project reproduces that comparison on two of the datasets, highD and NGSIM, and extends it in the direction of applied stochastic modelling: the fitted headway distribution is placed inside two stochastic processes (a renewal counting process for vehicle arrivals and a Markov chain for car-following regimes), and each model is verified against theory, simulation and data.

1.2 Why headway modelling matters

Three uses motivate a careful treatment. First, *capacity and flow*: the mean headway fixes the saturation flow of a lane. Second, *safety*: the lower tail of the distribution is the part that matters for rear-end conflicts, and a distribution that puts mass at implausibly short headways misstates it. Third, *simulation and control*: microscopic simulators and signal-timing procedures need a generator of headways or arrival counts, and the choice between a Poisson stream and a more regular renewal stream changes predicted queues and delays. These uses place different demands on the model (the bulk of the distribution, its lower tail, and the variability of counts), which is why the paper evaluates the same data with several criteria and not one.

1.3 Research objectives

1. Estimate the parameters of seven headway distributions on highD and NGSIM by Bayesian MCMC and compare their goodness of fit.
2. Reproduce the published goodness-of-fit results, and identify where and why they differ.
3. Construct a renewal counting process from the headways, derive its theoretical moments, and compare them with simulated sample paths, with a Poisson process, and with vehicle counts at a virtual detector reconstructed from the trajectories.
4. Construct a three-state regime Markov chain from within-trajectory transitions, derive its stationary distribution and recurrence times, and test the Markov property.
5. Examine the assumptions behind the estimation (likelihood, sampler convergence, goodness-of-fit metrics) and the stochastic models (independence of headways, the Markov property).

1.4 Research contributions

The contributions are as follows. (i) A reproduction of the headway-distribution comparison of [4] on highD and NGSIM, with an explicit account of the sample and metric definitions needed to reproduce it. (ii) A statistical audit of the estimation: the published likelihood does not condition on the truncation of the sample to $[0.5, 25]$ s, and re-estimating with the conditioned likelihood changes the highD ranking (Section 7.3.2); a non-converging posterior (shifted exponential on highD) is traced to the geometry of its likelihood and removed (Section 7.3.1); and the Wasserstein distance of the published study is shown to admit two non-equivalent readings (Section 4). (iii) A renewal-process and a regime-Markov-chain model built on these headways, with their theory, simulation and tests of their assumptions on real within-trajectory sequences. (iv) A reconstruction of detector inter-arrival times from the trajectories, which separates what the headway-driven renewal model predicts from what a detector would count (Section 7.3.4).

(v) A clear separation of what is verified mathematically, by simulation, and against real data (Section 7.3).

1.5 Organisation of the paper

Section 2 reviews the literature. Section 3 describes the datasets and preprocessing. Section 4 gives the mathematical background and Section 5 the proposed stochastic modelling framework. Section 6 describes the methodology, including the computational implementation. Section 7 presents the results, validation and discussion. Section 8 lists limitations and future work, and Section 9 concludes. The appendix contains source code only.

2 Literature Review

2.1 Historical development of traffic headway modelling

Probabilistic descriptions of road traffic begin with Adams [5], who treated the passage of vehicles at a point as a random series and described the intervals between consecutive vehicles by the negative exponential law. The link to the Poisson process is direct: if the number of vehicles in disjoint intervals is independent and Poisson distributed, the gaps between vehicles are independent and exponential, and conversely [2, 5, 6]. The Poisson–exponential pair is attractive because it has one parameter and is memoryless, but its density is largest at zero headway. Real drivers cannot follow at zero gap, because vehicles have length and drivers need time to react, and this inconsistency motivated modified forms. The literature reviewed in [4] and surveyed in [1] includes the shifted exponential, the hyper-exponential, the log-normal distribution [7], the hyperlang distribution [8] and the semi-Poisson model of Buckley [9]. Early studies assumed that all vehicles travel at free-flow speed and that headways are independent; as traffic data collection improved, studies of denser traffic relaxed both assumptions [4]. Cowan [10] gave a structural argument that has shaped later work: a headway is either a *tracking* headway, produced by a vehicle that follows its leader, or a *free* headway, produced by a vehicle that does not, and the composite models M1–M4 mix the two components. A distinction that runs through the later literature is between uninterrupted flow, where vehicles interact mainly with one another, and interrupted flow, where signals or stop signs break the stream [11]; headways are generally shorter under the former [4].

2.2 Classical probability distributions for headway modelling

Six families other than the proposed density recur in the headway literature and in the comparison of this paper (Table 2). The *negative exponential* is the Poisson benchmark above. Its *shifted* version places a minimum headway γ below which no headway occurs and keeps the memoryless tail, which makes it suitable for light traffic and unsuitable where the mode is away from the minimum. The *log-normal* distribution is right-skewed with positive support and was applied to headways early [7]; with a location shift it also excludes headways below a physical minimum. The *Weibull* distribution has a flexible hazard rate and contains the exponential as a special case [12]. The *gamma* distribution (an Erlang distribution when its shape is an integer) arises as the sum of independent exponential stages, which suggests a car-following interpretation; the hyperlang of [8] is a related generalisation. The *log-logistic* distribution has a closed-form distribution function and a heavier tail than the log-normal. The three-parameter *Burr* distribution generalises the log-logistic family (the log-logistic is its special case $\beta = 1$ in the parametrisation of Table 2) and spans a wide range of shapes. *Composite and mixture* models, such as Cowan's M3 and M4 models [10] and the mixed distribution for urban roads of Li et al. [13], combine a tracking and a free component and fit more flexibly, at the cost of more parameters and of a classification of

traffic states. The advantages of the single-component families are simplicity and well-known properties; their common limitation is that each describes a single regime, so that the best-fitting family changes with traffic conditions. Studies reviewed in [4] report log-normal fits for free-flowing traffic and log-logistic fits for congested traffic, Roy and Saha [14] compare candidate models for two-lane roads under mixed traffic, and Wang et al. [15] find that car-following-car and car-following-truck headways are described by log-logistic distributions while truck-following headways are better described by gamma distributions. Wu et al. [16] respond to this lack of a universal model by averaging several distributions with Bayesian model averaging; the finding that no single model describes all datasets motivates the search for one flexible, interpretable family.

2.3 Traffic heterogeneity and driver behaviour

Headway depends on at least four groups of factors. *Vehicle type*: Hoogendoorn and Bovy [17] estimate headway distributions per vehicle type, and Ye and Zhang [18] analyse vehicle-type-specific headways from freeway data. *Speed*: Chen et al. [19] show with NGSIM data that headway and spacing distributions differ across speed ranges, and Zou et al. [20] construct a bivariate distribution for freeway speed and headway. *Traffic condition and density*: headway distributions have been fitted to expressway, two-lane and heterogeneous traffic [14, 21, 22], and as reviewed in [4] the best-fitting family depends on the traffic condition. *Driver behaviour*: Taieb-Maimon and Shinar [23] study the minimum and the comfortable headways that drivers keep and how they relate to drivers' perception, which is the usual justification for a lower threshold of about 0.5 s. Automated vehicles add a further source of heterogeneity: Mahmassani [24] discusses the flow and operations considerations of traffic that mixes human-driven and automated vehicles, and Wen et al. [25] characterise the car-following behaviour of human drivers that follow automated vehicles. Datasets from automated vehicles, such as the Waymo open dataset [26], make research of this kind possible, and [4] includes two of them. The present study uses human-driven highway data only, so that heterogeneity enters through the pooling of vehicle types, drivers, lanes and speeds in each dataset and not through an explicit covariate.

2.4 Bayesian estimation in traffic modelling

Bayesian inference treats the parameters θ of a distribution as random, with a prior $p(\theta)$ that encodes what is known beforehand, and combines it with the likelihood of the data to give a posterior $p(\theta \mid x) \propto p(x \mid \theta)p(\theta)$ [27]. For distributions with several parameters this posterior is rarely available in closed form and is explored by Markov chain Monte Carlo, in particular the Metropolis–Hastings algorithm [28, 29]. A random-walk proposal is efficient when its scale is tuned: for high-dimensional targets, the optimal acceptance rate is about 23% [30], and proposals can be adapted during a warmup phase [31]. Convergence is judged by running several chains from dispersed starting points and comparing between-chain with within-chain variance [32, 33]. The advantages for headway modelling are that the posterior quantifies parameter uncertainty, that priors regularise poorly identified parameters, and that all candidate distributions are estimated in one framework with the same machinery. Wu et al. [16] use Bayesian model averaging to combine headway distributions, and Leungbootnak et al. [4] estimate each distribution by Metropolis–Hastings and compare the posterior means across datasets.

2.5 Renewal processes in traffic flow

A renewal process is a counting process whose inter-event times are independent and identically distributed [6, 34]. Taking the headways as inter-arrival times gives a model of the number of vehicles that pass a point, with the Poisson process as the special case of exponential headways.

The renewal formulation is natural for traffic because its inter-arrival distribution is free to be more regular than exponential: car following suppresses both very short and very long gaps, so that counts in a fixed interval vary less than a Poisson process would predict. The Poisson assumption may therefore overstate the variability of arrival counts, and it ignores the minimum headway that vehicle length and reaction time impose. Whether real headways are independent is a separate question, because a driver's gap to the leader changes gradually, and the early studies cited above that assumed independence did so for convenience [4]. The independence assumption is therefore tested and not assumed in this study (Sections 7.3.3 and 7.3.4).

2.6 Markov chains in traffic regime modelling

A finite Markov chain describes a system that moves between a small set of states with probabilities that depend only on the current state [35]. Chen et al. [19] developed a Markov model for the headway and spacing distribution of road traffic. Regime models of this kind need a definition of the states, an estimate of the transition matrix from trajectories, and a check that the Markov property is adequate. Their useful outputs are the stationary distribution, which gives the long-run share of time in each regime, the mean recurrence time of each regime, and the persistence of each regime, given by the diagonal of the transition matrix. A chain observed at one-second intervals along a single vehicle's trajectory treats the regime as a state of the driver–vehicle pair, which is the interpretation used here.

2.7 Research gap and motivation

The study being reproduced [4] compares headway *distributions* on five datasets. It does not place the fitted distribution inside a stochastic process, does not examine whether successive headways of a vehicle are independent, and does not report arrival-count behaviour or regime dynamics. It also leaves some details of its estimation and metrics open: the treatment of truncation in the likelihood, the definition of the vectors in its Wasserstein distance, and the NGSIM sample. Comparing the proposed distribution with classical alternatives on highD and NGSIM, while adding a renewal and a Markov model and verifying each of them, addresses these points. It shows how far the published conclusions carry over (they depend on the likelihood used) and how far a headway-driven renewal model describes what a detector would actually count, which a distribution comparison alone cannot show.

3 Dataset Description and Preprocessing

3.1 The highD dataset

highD [36] contains trajectories of road users recorded by a drone above German highways. The release used here comprises 60 recordings at 25 Hz. For each vehicle and frame, the tracking file gives the position, velocity, lane and the identifier of the preceding vehicle in the same lane (`precedingId`, zero if there is none), together with the time headway to it (`thw`); a metadata file gives the frame rate. highD describes uninterrupted highway flow of human-driven vehicles.

3.2 The NGSIM dataset

The Next Generation Simulation program of the U.S. Federal Highway Administration [37] released vehicle trajectories extracted from video of several roads. We use the I-80 data, recorded at 10 Hz. Each record gives the vehicle identifier, frame, lane, the preceding vehicle (`Preceding`, zero if none), the time headway (`Time_Headway`), the position along the road (`Local_Y`) and the time stamp (`Global_Time`). The study being reproduced [4] used the reconstructed I-80

trajectories of Montanino and Punzo [38]; the data used here is the original Data.gov export, which is not reconstructed.

3.3 Sampling, filtering and headway extraction

Following [4], headways are extracted as follows. (1) One sample per second is kept for every following vehicle (every 25th frame in highD, every 10th in NGSIM), so that the sampling rate is the same in both datasets. (2) Only records with a preceding vehicle in the same lane are kept. (3) Headways below $\alpha = 0.5$ s, which are generally regarded as below the minimum safe headway [23], and above $T = 25$ s, which do not represent car following, are discarded. The headway of a record is the dataset's own time-headway field: the time gap between a follower and its leader at one instant, not the time between successive vehicles passing a fixed point. This distinction matters for the renewal model (Section 5.3). The retained headways are ordered by vehicle and then time, and every stretch of consecutive seconds of one vehicle is labelled as a *trajectory run*. A new run starts whenever the vehicle changes or a second is missing, which happens when the vehicle has no valid leader for a while. Runs are used to avoid counting transitions or correlations between different vehicles (Section 6.8). Samples are not further cleaned: values outside the window are removed and no other outlier rule is applied.

3.4 Dataset selection rationale

The two datasets differ in the way that matters for headway: highD records free-flowing highway traffic with short headways, while NGSIM I-80 has longer and more varied headways (Table 1). Both are human-driven, publicly available and used by the study being reproduced. The NGSIM I-80 export contains two recording periods, separated by gaps of more than ten minutes in the time stamp. The first period gives 117,068 headways with a mean of 3.08 s; the second gives 292,831 headways with a mean of 4.04 s (both periods together: 409,899, mean 3.77 s). The published study does not say which period it used. The fitted NGSIM parameters in its Table 3 imply mean headways of 2.77–3.04 s for the 5 of its seven distributions that satisfy our parameter bounds (the mean of the window-conditioned model), which is close to the first period and far from the second. We therefore use the first period; this is an inference from the published numbers and not something the paper states, and the published NGSIM sample is not reproduced exactly (Section 7.3.6).

Table 1: Characteristics of the two datasets and of the headway samples used. Headways are in seconds; Std is the standard deviation. A trajectory run is a stretch of consecutive one-second samples of one follower (Section 3.3).

	highD	NGSIM (I-80)
Provider	RWTH Aachen University [36]	U.S. Federal Highway Administration [37]
Region and sensor	German highways; drone camera	I-80, California; fixed video cameras
Native frame rate	25 Hz	10 Hz
Flow regime	Uninterrupted, human-driven	Freeway, human-driven
Selection used	all 60 recordings	first recording period
Leader and headway fields	`precedingId`, `thw`	`Preceding`, `Time_Headway`
Sampling	one sample per second per follower	one sample per second per follower
Retained headways	1,225,460	117,068
Mean / median / Std (s)	2.21 / 1.66 / 1.73	3.08 / 2.56 / 2.08
Min / max (s)	0.50 / 24.96	0.50 / 24.98
Trajectory runs; mean run length (s)	112,841; 10.9	3,207; 36.5
Within-run transitions	1,112,619	113,861

4 Mathematical Background

4.1 Probability distributions and truncation

Seven distributions are compared (Table 2): the proposed distribution of Section 5.2 and six standard ones, with the densities and the priors of [4]. Headways are observed only in the window $[\alpha, T] = [0.5, 25]$ s. A distribution with density f and distribution function F *conditioned on the window* has density and distribution function

$$f_{\mathrm{w}}(t) = \frac{f(t)}{F(T) - F(\alpha)}, \qquad F_{\mathrm{w}}(t) = \frac{F(t) - F(\alpha)}{F(T) - F(\alpha)}, \qquad \alpha \le t \le T. \tag{4.1}$$

The simulation of headways and every goodness-of-fit measure below use F_{w}.

Table 2: The seven distributions and the priors of [4]. $\mathcal{N}(m, s)$ has mean m and standard deviation s; $\Gamma(k, r)$ has shape k and rate r; $t_{\min}$ is the smallest observed headway.

Distribution	Density	Priors
Proposed	$\frac{b^{\lvert t-a\rvert}}{Z(a,b)}$, $t \ge \alpha$	$a \sim \mathcal{N}(0, 10)$, $b \sim U(0, 1)$
Shifted log-normal	$\frac{1}{(t-\gamma)\sigma\sqrt{2\pi}} \exp\Big\{ - \frac{[\ln(t-\gamma) - \mu]^2}{2\sigma^2} \Big\}$, $t > \gamma$	$\mu \sim \mathcal{N}(0, 10)$, $\sigma \sim \Gamma(0.5, 0.5)$, $\gamma \sim U(-10, t_{\min})$
Weibull	$\frac{\alpha t^{\alpha-1}}{\beta^{\alpha}} e^{-(t/\beta)^{\alpha}}$	$\alpha, \beta \sim \Gamma(0.5, 0.5)$
Log-logistic	$\frac{(\alpha/\beta)(t/\beta)^{\alpha-1}}{[1 + (t/\beta)^{\alpha}]^2}$	$\alpha, \beta \sim \Gamma(0.5, 0.5)$
Gamma	$\frac{\beta^{\alpha}}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}$	$\alpha, \beta \sim \Gamma(0.5, 0.5)$
Burr	$\frac{\alpha\beta}{\lambda} (t/\lambda)^{\alpha-1} \big[1 + (t/\lambda)^{\alpha}\big]^{-\beta-1}$	$\alpha, \beta, \lambda \sim \Gamma(0.5, 0.5)$
Shifted exponential	$\lambda e^{-\lambda(t-\gamma)}$, $t \ge \gamma$	$\lambda \sim \Gamma(0.5, 0.5)$, $\gamma \sim U(-10, t_{\min})$

4.2 Bayesian inference and the Metropolis–Hastings algorithm

Let $\mathbf{n} = (n_1, \dots, n_{49})$ be the counts of headways in the 49 bins of width 0.5 s that tile $[0.5, 25]$ s, and $p_k(\theta)$ the model probability of bin k. The posterior is

$$\log p(\theta \mid \mathbf{n}) = \log p(\theta) + \sum_{k=1}^{49} n_k \log p_k(\theta) + \text{const.} \tag{4.2}$$

Two definitions of p_k are used (Section 6.4). The Metropolis–Hastings algorithm [28, 29] generates a Markov chain $\theta^{(0)}, \theta^{(1)}, \dots$ whose stationary distribution is the posterior. From θ a candidate $\theta' = \theta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, s^2\Sigma)$ is proposed and accepted with probability

$$\rho(\theta, \theta') = \min\Big\{1,\ \frac{p(\theta' \mid \mathbf{n})}{p(\theta \mid \mathbf{n})}\Big\}, \tag{4.3}$$

and otherwise the chain stays at θ. The proposal is symmetric, so no proposal ratio appears. The scale s and covariance Σ affect only the efficiency. For random-walk proposals the optimal acceptance rate is near 23% [30].

4.3 Convergence diagnostics

For C chains of N draws, the split $\hat{R}$ [27, 32, 33] splits every chain in two, giving $m = 2C$ sequences of length $n = N/2$ with means $\bar{\theta}_j$, and compares the between-sequence variance B with the within-sequence variance W:

$$\hat{R} = \sqrt{\frac{\frac{n-1}{n}W + \frac{1}{n}B}{W}}, \qquad B = \frac{n}{m-1}\sum_{j=1}^{m}(\bar{\theta}_j - \bar{\theta})^2. \tag{4.4}$$

Values close to 1 indicate that the sequences agree; we flag $\hat{R} \geq 1.05$. This is the classical split statistic, not the rank-normalised variant of [33]. The effective sample size (ESS), estimated from the autocorrelation of each chain, measures how many independent draws the chain is worth.

4.4 Renewal processes and the Poisson process

Let $H_1, H_2, \dots$ be i.i.d. positive random variables with mean μ and variance σ^2 and let $S_n = H_1 + \cdots + H_n$. The counting process $N(t) = \max\{n : S_n \leq t\}$ is a renewal process, with $\{N(t) \geq n\} = \{S_n \leq t\}$. Its renewal function $m(t) = \mathbb{E}N(t)$ satisfies $m(t)/t \to 1/\mu$ (the elementary renewal theorem), and for a non-lattice distribution with finite variance [6, 34]

$$m(t) = \frac{t}{\mu} + \frac{\sigma^2 - \mu^2}{2\mu^2} + o(1), \qquad \operatorname{Var} N(t) \sim \frac{\sigma^2 t}{\mu^3}, \qquad N(t) \ \dot{\sim}\ \mathcal{N}\Big(\frac{t}{\mu},\ \frac{\sigma^2 t}{\mu^3}\Big). \tag{4.5}$$

If H is exponential then $N(t)$ is a Poisson process, with $\operatorname{Var} N(t) = \mathbb{E}N(t) = t/\mu$ for all t. In general

$$\frac{\operatorname{Var} N(t)}{\mathbb{E}N(t)} \ \longrightarrow\ \frac{\sigma^2}{\mu^2} =: \mathrm{CV}^2, \tag{4.6}$$

the squared coefficient of variation of the headway, also called the index of dispersion of the counts. A value below 1 means that arrivals are more regular than Poisson, and above 1 more clustered.

4.5 Finite-state Markov chains

A discrete-time Markov chain (X_n) on a finite state space S with transition matrix $P = (p_{ij})$ satisfies $\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots) = p_{ij}$, and its n-step transition matrix is $P^{(n)} = P^n$, the Chapman–Kolmogorov equation [35],

$$P^{(m+n)} = P^{(m)} P^{(n)}. \tag{4.7}$$

If the chain is irreducible and aperiodic, $P^n \to \mathbf{1}\pi$, where the stationary distribution is the solution of

$$\pi P = \pi, \qquad \sum_{i \in S} \pi_i = 1, \tag{4.8}$$

and the mean recurrence time of state i is $1/\pi_i$. The expected time spent in state i per visit is $1/(1 - p_{ii})$. The maximum-likelihood estimate of P from observed paths is $\hat{p}_{ij} = n_{ij}/n_{i\cdot}$, where n_{ij} is the number of observed transitions from i to j.

4.6 Goodness-of-fit metrics

Four measures from [4] are computed on the window-conditioned model. The *Kolmogorov–Smirnov* (KS) statistic [39]

$$D = \sup_x |Q(x) - P(x)| \tag{4.9}$$

compares the empirical distribution function Q of the data with the distribution function P either of the fitted model (one-sample test) or of a sample drawn from it (two-sample test, which is how the published study describes it). The *chi-square* statistic [40]

$$\chi^2 = \sum_i (O_i - E_i)^2 / E_i \tag{4.10}$$

is computed on the bin counts, with adjacent bins merged until each merged bin has an observed count of at least 5, and has (number of merged bins $-1-$ number of parameters) degrees of freedom. The *Kullback–Leibler divergence* of the observed bin probabilities p_k from the model bin probabilities q_k [41] is

$$D_{KL}(P \| Q) = \sum_k p_k \log \frac{p_k}{q_k}. \tag{4.11}$$

The *Wasserstein distance* of order 1 between two samples of equal size n is the mean absolute difference of their order statistics, $\frac{1}{n}\sum_{i=1}^{n} |X_{(i)} - Y_{(i)}|$ [42]; this is Eq. 17 of [4] with $p = 1$. That equation does not say what X and Y are, so two quantities that must not be confused are evaluated:

$$W_{\text{vec}} = \frac{1}{49} \sum_{i=1}^{49} |p_{(i)} - q_{(i)}|, \qquad \text{(sorted bin-probability vectors, unitless)} \tag{4.12}$$

$$W_{\text{time}} = \int \left| F_{\text{obs}}(t) - F_{\text{w}}(t) \right| dt, \qquad \text{(headway distributions on the time axis, seconds)} \tag{4.13}$$

where $p_{(i)}$ and $q_{(i)}$ are the observed and model bin probabilities in increasing order. W_{vec} treats the 49 probabilities as two samples and is blind to *where* on the time axis each bin lies. W_{time} is the conventional one-dimensional Wasserstein distance between the two binned headway distributions, evaluated with the mass of each bin placed at its centre. The two are not equivalent: W_{vec} is invariant to permuting the bins, W_{time} is not, and they can rank models differently (Section 7.2.3). The values printed in [4] (of order 10^{-3}) are only compatible in scale with W_{vec}, so that is the quantity compared with the paper; W_{time} is reported alongside it because it is the conventional distance.

5 Proposed Stochastic Modelling Framework

5.1 Terminology

Five terms are kept apart throughout (Table 3). The distinction between a *time headway* and a *detector inter-arrival time* is central: the first is measured between two vehicles at one instant, the second between two vehicles passing one place.

Table 3: Terminology used in this paper.

Term	Meaning here
Time headway H	Time gap between a following vehicle and its leader in the same lane at one instant, as reported by the datasets (`thw`, `Time_Headway`).
Detector inter-arrival time	Time between two successive vehicles of a lane crossing a fixed cross-section. Reconstructed from the trajectories in Section 7.3.4.
Renewal inter-arrival distribution	The distribution assigned to the i.i.d. gaps of the renewal process $N(t)$; in the headway-driven model, the time-headway distribution.
Vehicle trajectory	The sequence of positions of one vehicle over time; a *run* is a stretch of consecutive seconds of it that enters the sample.
Traffic regime	One of three classes of time headway of a follower (close following, stable platoon, free flow), Section 5.4.

5.2 Bayesian headway distribution model

The proposed density is

$$f(t \mid a, b) = \frac{b^{|t-a|}}{Z(a,b)}, \qquad t \geq \alpha = 0.5 \text{ s}, \quad 0 < b < 1, \tag{5.1}$$

with parameters a (location of the mode, in seconds) and b (dimensionless dispersion). With $\kappa = -\ln b > 0$ it is $f \propto e^{-\kappa|t-a|}$, a Laplace (double-exponential) density centred at a and truncated to $t \geq \alpha$. For $a > \alpha$ the density has its mode at a, and integrating separately on $[\alpha, a]$ and $[a, \infty)$ gives the normalising constant (5.2) and distribution function (5.3)

$$Z = \frac{2 - b^{a-\alpha}}{\kappa}, \tag{5.2}$$

$$F(t) = \begin{cases} \dfrac{b^{a-t} - b^{a-\alpha}}{2 - b^{a-\alpha}}, & \alpha \leq t \leq a, \\[2ex] \dfrac{2 - b^{a-\alpha} - b^{t-a}}{2 - b^{a-\alpha}}, & t \geq a. \end{cases} \tag{5.3}$$

For $a \leq \alpha$ the density decreases on $[\alpha, \infty)$, with $Z = b^{\alpha-a}/\kappa$ and $F(t) = 1 - b^{t-\alpha}$. Equation (5.3) agrees with the closed form of [4] and is checked against numerical integration in the test suite (Section 7.3). Inverting it gives, with $c = b^{a-\alpha}$ and $u \sim U(0,1)$, the inverse-transform expression

$$t = \begin{cases} a - \ln\big(u(2-c) + c\big)/\ln b, & u \leq F(a), \\ a + \ln\big((1-u)(2-c)\big)/\ln b, & u > F(a), \end{cases} \tag{5.4}$$

which is stated for completeness; the code instead inverts the window-conditioned distribution function on a fine grid, the same way for all seven distributions (Section 6.9). The parameters have a direct reading: a is the most frequent headway (the mode), and κ is the exponential rate at which the density falls on both sides of it. Because the decay rate is the same on both sides of a, the density is symmetric about its mode apart from the truncation at α, which constrains the shapes it can take (Section 7.2.4). With b near 1 the density is nearly flat, in agreement with the interpretation of [4].

5.3 Renewal arrival process

Definition. The headways $H_1, H_2, \dots$ of a lane are modelled as i.i.d. draws from a headway distribution (the data or a fitted distribution), and the number of vehicles that have passed a point by time $t \geq 0$ is the renewal counting process $N(t)$ of Section 4, with state space $\{0, 1, 2, \dots\}$ and continuous time. The theory of Eqs. (4.5)–(4.6) then gives the expected number of arrivals, its variance and its distribution.

What this model assumes. The model is *headway-driven*: it uses the time headway between a follower and its leader as the inter-arrival time. The two coincide when vehicles keep a constant speed between the leader's and the follower's passage of a point, but not in general, and the datasets report H at one instant, not detector crossing times. The second assumption is that the headways are independent. Both assumptions are tested: the independence on the one-second headways (Section 7.3.3), and the headway-versus-inter-arrival distinction by reconstructing detector crossing times from the trajectories (Section 7.3.4). The detector-based reconstruction is feasible with the available fields (position, frame or time stamp, lane), and it is used as a validation of the renewal model; the headway-driven process remains the model of record because it is the one that is linked to the fitted headway distributions.

5.4 Markov chain for traffic regimes

State space and time. Each retained one-second sample of a follower is assigned a regime

$$X_n = \begin{cases} \text{close following} & 0.5 \leq h < 1.5 \text{ s}, \\ \text{stable platoon} & 1.5 \leq h < 4 \text{ s}, \\ \text{free flow} & 4 \leq h \leq 25 \text{ s}, \end{cases} \tag{5.5}$$

where n indexes consecutive seconds of one vehicle's trajectory. The sequence (X_n) is modelled as a time-homogeneous, first-order Markov chain on $S = \{\text{close, platoon, free}\}$ with discrete time. The thresholds are modelling choices, not estimated quantities, and "free flow" here means only a headway above 4 s to a leader that is still present, since samples without a leader within 25 s are discarded.

Estimation and quantities. P is estimated by transition counts, $\hat{p}_{ij} = n_{ij}/n_{i\cdot}$, where only pairs of consecutive samples in the *same trajectory run* are counted (Section 6.8). The chain gives the stationary distribution (4.8), the mean recurrence times $1/\pi_i$, the mean sojourn time $1/(1-p_{ii})$ and the transient distribution $\pi(n) = \pi(0)P^n$. A mean recurrence time is the expected number of seconds between successive visits of a trajectory to the regime, assuming that the chain is stationary.

6 Methodology

6.1 Overall research workflow

The workflow has four stages (Figure 1): data are loaded and reduced to one-second headways; seven distributions are fitted by MCMC under two likelihoods and evaluated; the renewal process and the Markov chain are built from the headways; and each stage is verified, with a detector-based test of the renewal model (Section 7.3). All stages are run by scripts and write their tables and figures to disk.

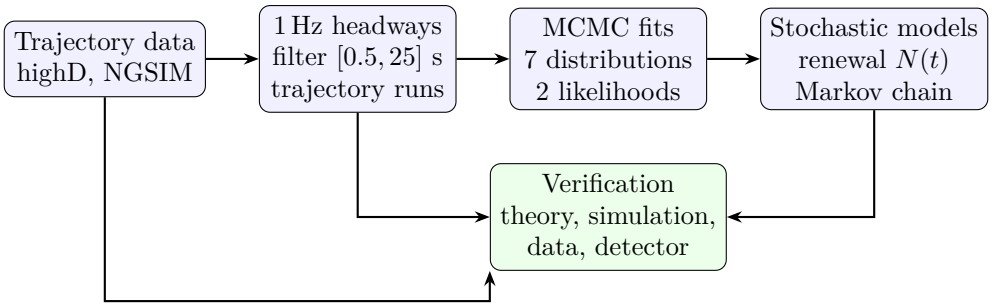


Figure 1: Workflow of the study.

6.2 Data acquisition and preprocessing

The loaders read each recording or the NGSIM CSV and apply the extraction of Section 3: one-second sampling, requirement of a preceding vehicle, and the window $[0.5, 25]$ s. NGSIM records are selected by location (I-80) and by recording period, and are sorted by vehicle and time, because the raw export is not in that order. The loaders also return the trajectory-run label of every headway. The highD recordings are processed in parallel, one task per recording. The raw datasets are the highD release (60 recordings) and the NGSIM Data.gov export; they are not redistributed with this paper.

6.3 Headway distribution selection

The candidate set is the seven distributions of Table 2, fixed in advance by the study being reproduced. Selection is by the goodness-of-fit measures of Section 4.6, and no model is chosen by a test of significance, since all of them are rejected on samples of this size (Section 7.2.3). Models are ranked primarily by the KL divergence, which is the criterion that the published study emphasises, and the two Wasserstein distances and the KS statistic are read as checks.

6.4 Bayesian parameter estimation

Each distribution is fitted separately to each dataset, using the log-posterior of Eq. (4.2) with the priors of Table 2 and the 49 bin counts. This is the likelihood of the binned data: it has the same cost per step whatever the sample size and discards the position of a headway inside its 0.5 s bin. Two definitions of the bin probabilities p_k are implemented and both are run:

- the *paper likelihood* uses each distribution's own density, $p_k = F(e_{k+1}) - F(e_k)$ for the bin $[e_k, e_{k+1})$, so that $\sum_k p_k < 1$ because mass outside $[0.5, 25]$ s is lost. This is the method of [4] and is used to reproduce it;
- the *truncated* (window-conditioned) likelihood uses $p_k = [F(e_{k+1}) - F(e_k)]/[F(T) - F(\alpha)]$, which is Eq. (4.1) integrated over the bin, sums to one, and is the exact multinomial likelihood for a sample truncated to the window. It is the correct likelihood for these data and is used for all corrected results.

The two coincide for the proposed distribution, which has no mass below α and a negligible one above 25 s. They differ for distributions on $(0, \infty)$ whenever they put mass below 0.5 s (Section 7.3.2). Under the truncated likelihood the threshold γ of the shifted exponential is not identifiable, because the conditioned distribution does not depend on it (for $\gamma < \alpha$, F_{w} is the exponential distribution function conditioned on the window); its posterior is its prior, and it is reported as such. All other parameters are identified.

6.5 Metropolis–Hastings sampling procedure

Two chains of 10 000 iterations are run for each fit, the first 5 000 being warmup, as in [4]. The procedure is:

1. **Initialise.** Chain 1 starts at the maximum a posteriori (MAP) point found by the Nelder–Mead simplex; chain 2 starts from this point scaled by 1.05 if that point is feasible, and from the MAP point otherwise.
2. **Propose.** $\theta' = \theta + s\,L\,z$ with $z \sim \mathcal{N}(0, I)$ and $LL^\top = \Sigma$; initially $\Sigma = \mathrm{diag}\{(0.1\max(|\theta_i|, 0.5))^2\}$ and $s = 2.38/\sqrt{d}$ for d parameters.
3. **Accept or reject** with the probability of Eq. (4.3).
4. **Adapt during warmup only.** Every 100 iterations s is multiplied by $\mathrm{clip}\{\exp[0.8(r - 0.234)], 0.5, 2\}$, where r is the recent acceptance rate, and every 500 iterations Σ is re-estimated from the last 1 000 draws. Nothing is adapted after warmup, which preserves the stationary distribution.
5. **Estimate.** The point estimate is the mean of the 10 000 retained draws (2 chains of 5 000).

The shifted exponential's threshold $\gamma < t_{\min}$ is sampled through $v = \log(t_{\min} - \gamma)$, with the Jacobian e^v added to the log-density, which is the exact change of variables for the uniform prior on γ; the MAP start is polished in the same coordinates. The reason for this reparametrisation, and its effect, are in Section 7.3.1.

6.6 Goodness-of-fit evaluation

For each fit, the window-conditioned model is evaluated with the measures of Section 4.6: the one-sample KS statistic against F_{w} and the two-sample KS statistic of the data against a sample of $\min(n, 50\,000)$ draws from the fitted model, the chi-square statistic on the merged bins, D_{KL} of Eq. (4.11), and W_{vec} and W_{time} of Eqs. (4.12)–(4.13). The chi-square degrees of freedom use the nominal number of parameters of each distribution (including γ of the shifted exponential).

6.7 Renewal process construction

For each dataset the renewal process is built on the empirical headways of the sample. The mean μ and variance σ^2 of the headway give $m(t)$, $\mathrm{Var}\,N(t)$ and the dispersion σ^2/μ^2 of Eqs. (4.5)–(4.6), and the lane flow $3600/\mu$ vehicles per hour. This is the headway-driven process of Section 5.3. A second construction uses detector inter-arrival times (Section 6.10).

6.8 Markov chain construction

Headways are mapped to regimes by Eq. (5.5). The transition matrix is estimated from *within-run* pairs only: the pair (x_i, x_{i+k}) is used only if both samples carry the same trajectory-run label. Since a run is a stretch of consecutive seconds of one vehicle, this excludes every transition between two vehicles and every transition across a gap in time. For $k > 1$ the same rule gives the empirical k-step transition frequencies used in the Chapman–Kolmogorov check, and the autocorrelation of headways at lag k uses the same pairs. The stationary distribution is obtained by solving Eq. (4.8) by least squares.

6.9 Simulation design

Renewal paths. For $M = 2\,000$ paths, headways are drawn independently with replacement from the sample and cumulated to arrival times S_n; $N(t)$ is the number of arrivals up to t, at the

horizons $t = 30, 60, 120, 300$ s. Five paths are plotted. For the Markov chain, sample paths are generated from the estimated P and the transient distribution $\pi(n)$ is computed from the initial state "close following". All random numbers come from seeded generators (seed 42). Because headways are resampled independently, these paths have no serial dependence by construction.

Fitted-model sampling. Headways from a fitted distribution are drawn by inverting the window-conditioned distribution function on a grid of 50 001 points, for all seven distributions.

6.10 Detector inter-arrival times and count dispersion

A virtual loop detector is placed at a fixed longitudinal coordinate x_0. For a vehicle with consecutive samples (s_j, x_j) and (s_{j+1}, x_{j+1}) on opposite sides of x_0, the crossing time is obtained by linear interpolation

$$\tau = s_j + \frac{x_0 - x_j}{x_{j+1} - x_j}\,(s_{j+1} - s_j), \tag{6.1}$$

each vehicle being counted once, at its first crossing, and in the lane that it occupies at the crossing. The inter-arrival times are the differences of successive crossing times of one lane in one recording (or period); no difference is taken across lanes or recordings. In highD, x_0 is the midpoint of the range of vehicle-centre positions in the recording (x plus half the bounding-box length), time is frame divided by 25. In NGSIM, $x_0 = 800$ ft along `Local_Y`, time is `Global_Time`, only the first recording period and the mainline lanes 1–6 are used (the on-ramp lane 7 is excluded). From these series three quantities are computed: the moments of the inter-arrival times; their autocorrelation at lags 1–5, with each lane-recording series centred on its own mean so that differences in flow between lanes do not appear as dependence; and the index of dispersion of the vehicle count in consecutive windows of $w = 30, 60, 120$ s,

$$I_w = \frac{\operatorname{Var} N_w}{\mathbb{E} N_w} \qquad \text{against} \qquad \frac{\sigma_u^2}{\mu_u^2}\ \text{(renewal)} \qquad \text{and} \qquad 1\ \text{(Poisson)}, \tag{6.2}$$

where μ_u, σ_u^2 are the mean and variance of the inter-arrival times of the same lane-recording series u, and series are averaged with weights equal to their number of windows. Series with fewer than five windows or ten gaps are skipped. A flow that drifts during a recording inflates I_w above the stationary renewal value, so the comparison is a bound and not an exact test.

6.11 Statistical validation strategy

Three kinds of validation are kept distinct, because agreement of a simulation with its own theory does not validate the model.

1. **Mathematical verification**: the closed forms are correct. Automated tests check the distribution function against numerical integration, that every density integrates to one, and that the derivative of each distribution function is its density.
2. **Simulation verification**: the code does what the mathematics says. Automated tests check that the sampler recovers the known parameters of a distribution from a sample drawn from it, and the simulated renewal moments are compared with Eq. (4.5). Because the simulated paths are generated under the assumptions of the theory, agreement shows only that the simulation implements the theory.
3. **Empirical validation**: the model describes the real data. The fitted distributions are compared with the observed histograms by the metrics above; the assumptions of the renewal and Markov models are tested on the real sequences (autocorrelation, Chapman–Kolmogorov, first- against second-order, detector counts); and the results are compared with the published ones.

6.12 Computational implementation

The implementation is a Python package, `headway/`, with a driver script, `run.py`. It uses NumPy, SciPy, pandas and Matplotlib (versions pinned in `uv.lock`); the Metropolis–Hastings sampler is written from scratch. The modules are organised by role:

- `distributions.py`: the seven densities and distribution functions, the priors, and the conditioned quantities of Eq. (4.1);
- `mcmc.py` and `fit.py`: the sampler, the binned posterior with both likelihoods, the MAP start, and the threshold reparametrisation;
- `metrics.py` and `diagnostics.py`: the goodness-of-fit measures, split $\hat{R}$ and the effective sample size;
- `datasets/`: the highD and NGSIM loaders and the trajectory-run labelling;
- `stochastic_process.py`, `simulation.py` and `validation.py`: the renewal process and Markov chain, headway simulation, and the quantitative checks;
- `detector.py`: virtual-detector inter-arrival times and the count-dispersion test;
- `plots.py`, `make_report_assets.py` and `make_code_appendix.py`: the figures, and the generation of every number, table and listing of this paper from the output files.

Parallel processing. Independent chains and independent recordings are run on a shared process pool (`parallel.py`). Each worker process holds its own copy of the data it is loading; with the default of one worker per core the 32 GB Windows machine used here ran out of virtual memory, so the reported runs used six workers.

Automated result generation. `run.py` fits all seven distributions to both datasets under the chosen likelihood, evaluates them, runs the stochastic analyses and writes CSV tables and figures; it was run once per likelihood (`-likelihood truncated` and `paper`), and `python -m headway.detector` produced the detector analysis. `make_report_assets.py` reads these files and writes the macros (`results.tex`) and table bodies that this paper includes, so that no number in the text is typed by hand. The MCMC itself takes 14.5 s on highD and 16.3 s on NGSIM for all seven distributions (two chains each, one run); reading the raw trajectory files is the dominant cost and was not timed separately.

Testing. The automated test suite (52 tests, all passing) covers the distributions, the sampler, the loaders, the metrics, the Markov and renewal processes, the within-run bookkeeping, the shifted-exponential convergence, the truncated likelihood and the detector reconstruction.

Reproducibility. Every random generator is seeded (seed 42; chain c uses the generator seeded by $(42, c, 100)$), dependencies are pinned, and a test checks that the parallel fit equals the serial one, so results do not depend on the number of workers. The listings of the appendix are extracted automatically from the source files. The repository was run on Python 3.13. The code was run with one seed only: the Monte Carlo variability of the MCMC estimates is described by the posterior standard deviations and ESS of Table 14, and that of the simulated counts by Section 7.1.

7 Results and Discussion

7.1 Simulation results

7.1.1 Renewal arrival process

Figure 2 shows five simulated sample paths of the arrival count $N(t)$ for each dataset, with the renewal function $m(t)$ of Eq. (4.5). The paths scatter around $m(t)$. Table 4 compares 2 000 simulated paths with renewal theory. The simulated means agree with $m(t)$ to within 0.2% on highD and 0.2% on NGSIM at all four horizons. The simulated variances agree to within 3.8% on highD (largest at $t = 300$ s) and 7.0% on NGSIM (largest at $t = 30$ s). The sampling error of a variance estimated from 2 000 paths is about 3% (relative standard error $\sqrt{2/1999}$ for approximately normal counts), and Eq. (4.5) is an asymptotic result that is least accurate at the shortest horizon with only about ten arrivals, so these differences are consistent with Monte Carlo and asymptotic error. This is a verification that the simulation implements Eq. (4.5), not a validation of the model against data.

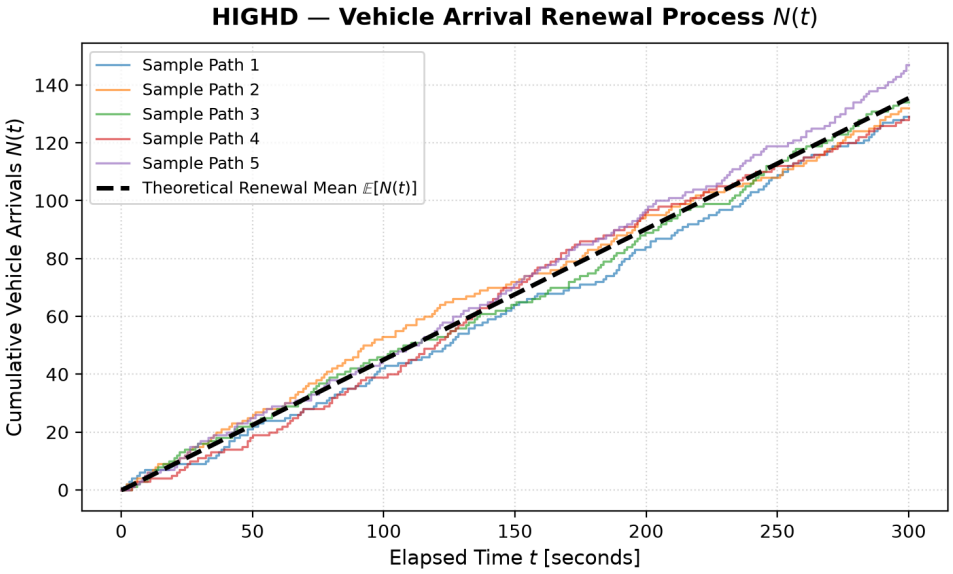
Table 4: Simulated renewal counts (2 000 paths) against renewal theory, Eq. (4.5). "Var if Poisson" is the variance of a Poisson process of the same rate $1/\mu$; errors are relative to theory. $\mathbb{E}N$: expected count; $\mathrm{Var}\,N$: variance of the count.

Dataset	t (s)	$\mathbb{E}N$ sim	$\mathbb{E}N$ theory	$\mathrm{Var}\,N$ sim	$\mathrm{Var}\,N$ theory	Var if Poisson	mean err	var err
highD	30	13.38	13.37	8.41	8.28	13.56	0.10%	1.6%
	60	26.86	26.93	16.86	16.56	27.12	0.23%	1.8%
	120	54.02	54.05	33.11	33.13	54.24	0.05%	0.0%
	300	135.57	135.41	85.94	82.81	135.60	0.12%	3.8%
NGSIM	30	9.47	9.45	4.12	4.43	9.73	0.18%	7.0%
	60	19.16	19.18	8.50	8.86	19.45	0.12%	4.1%
	120	38.72	38.63	17.20	17.71	38.90	0.24%	2.9%
	300	96.95	96.98	46.37	44.28	97.26	0.04%	4.7%

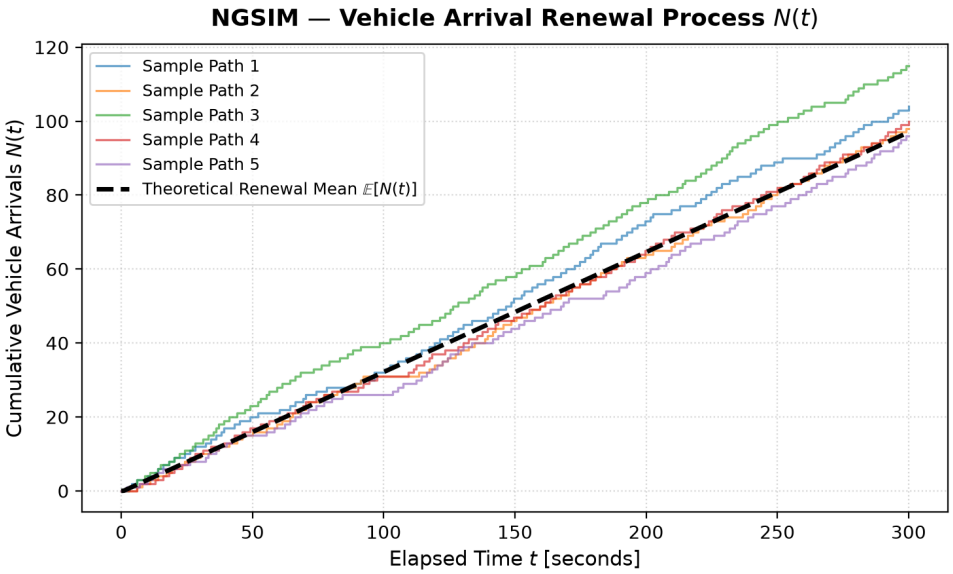
The variance of the count is far below that of a Poisson process: at $t = 300$ s it is 85.94 on highD against 135.60 for a Poisson process of the same rate (Table 4), and 46.37 on NGSIM against 97.26. The variance-to-mean ratio of Eq. (4.6) is 0.61 on highD and 0.46 on NGSIM, where a Poisson process has 1. Figure 3 shows the whole distribution of $N(60)$. At $t = 60$ s the largest gap between the simulated count distribution function and the renewal-theory normal approximation is 0.022 on highD and 0.038 on NGSIM, while the gap to a Poisson distribution is 0.071 and 0.127; the normal approximation of Eq. (4.5) describes the simulated counts much better than the Poisson distribution does. The renewal rates are 0.4520 veh/s on highD and 0.3242 veh/s on NGSIM, i.e. 1,627 and 1,167 veh/h/lane. These are the reciprocals of the mean headways of followers who have a leader, so they describe car-following states and are not measured detector flows (Section 7.3.4).

7.1.2 Regime Markov chain

Table 5 gives the transition matrix, the stationary distribution and the mean recurrence times of the regime chain of Section 5.4, estimated from within-run transitions only (1,112,619 on highD, 113,861 on NGSIM), and Figure 4 shows the chains as state-transition diagrams. Both chains are persistent, the highD chain much more than the NGSIM chain: the expected sojourn $1/(1 - p_{ii})$ is 33.7, 25.8 and 19.3 s for close following, platoon and free flow on highD, and 3.9, 12.5 and 4.9 s on NGSIM. Observed runs last only 10.9 s on average on highD and 36.5 s on NGSIM because a vehicle is observed only while it crosses the camera field and has a leader,

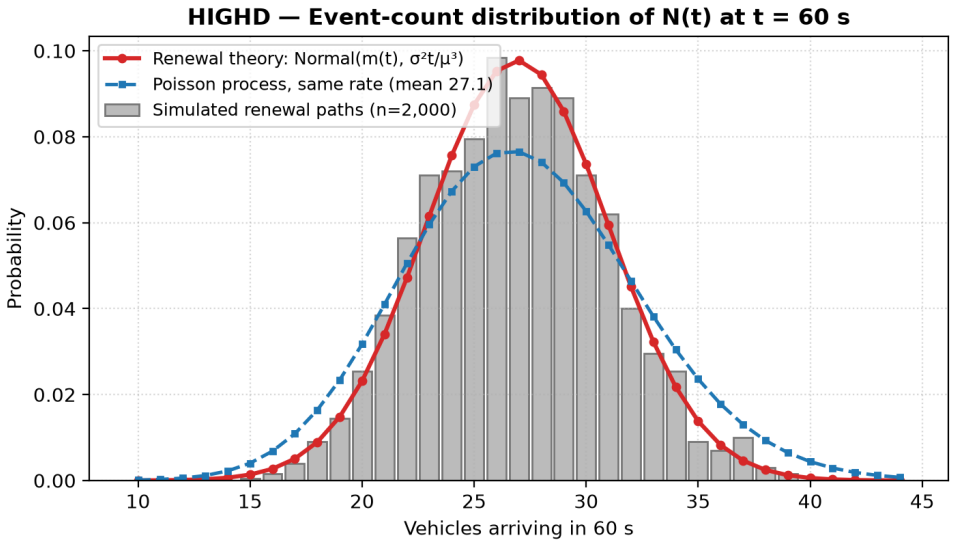


(a) highD

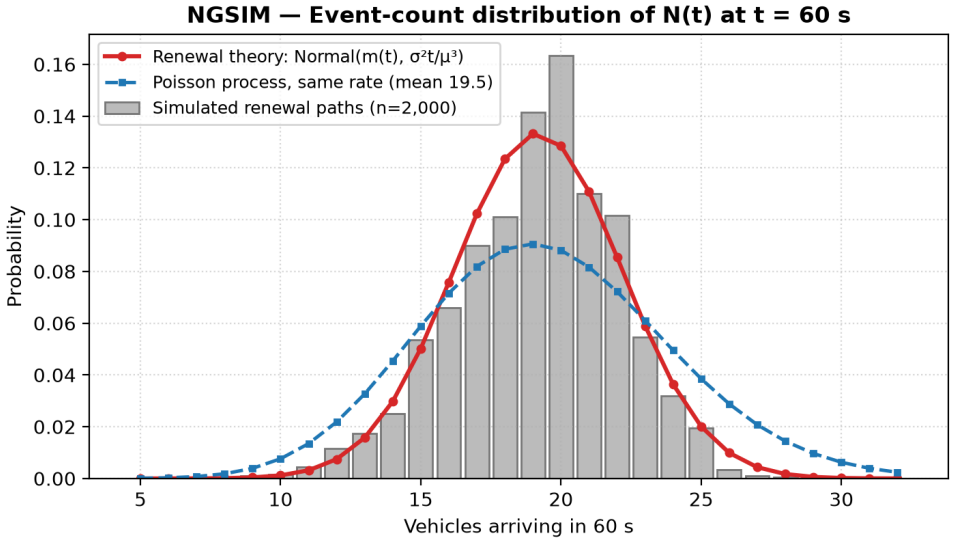


(b) NGSIM

Figure 2: Five simulated sample paths of the vehicle arrival count $N(t)$ (cumulative vehicles, t in seconds) of the headway-driven renewal process, and the theoretical renewal function $m(t)$ (dashed).



(a) highD



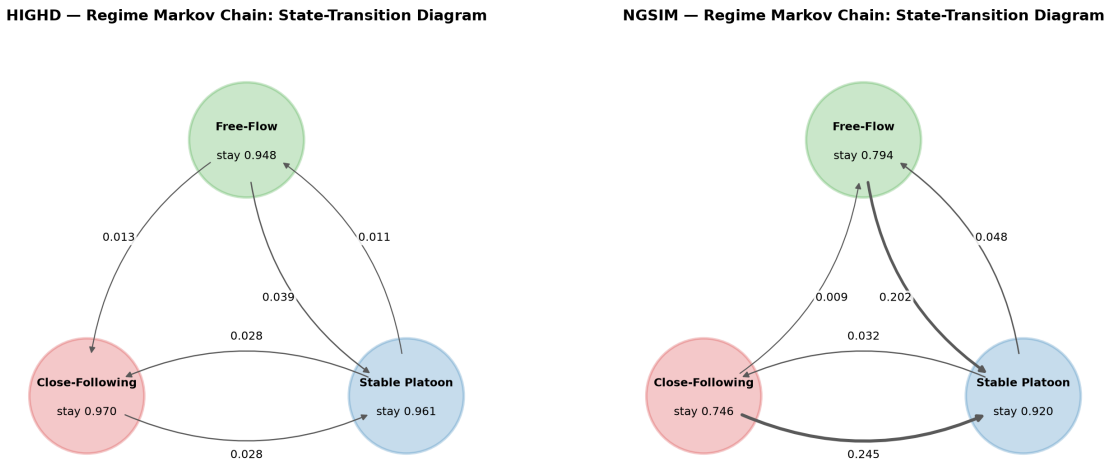
(b) NGSIM

Figure 3: Distribution of the arrival count $N(60\text{ s})$: simulated renewal paths (bars), the renewal-theory normal approximation with continuity correction (red) and a Poisson process of the same rate (blue).

so sojourns are censored by the observation window and the sojourn times are model-implied expectations, not directly observed durations. The two datasets differ strongly. On highD, the chain spends 0.454 of the time in close following and 0.435 in stable platoon, and only 0.111 in free flow; on NGSIM the platoon regime dominates (0.730), close following is rare (0.096), and free flow accounts for 0.174. The mean recurrence times follow: a highD trajectory returns to close following on average every 2.20 steps (seconds), whereas an NGSIM trajectory returns to it only every 10.42. Figure 5 shows the transient evolution $\pi(n) = \pi(0)P^n$ from close following: the NGSIM chain is within total-variation distance 0.003 of its stationary distribution after 20 seconds, whereas the highD chain, whose regimes are more persistent, is at 0.182 after 20 seconds and 0.022 after 60 seconds, which is much longer than the mean observed run of 10.9 s.

Table 5: Estimated transition matrix P (rows: current regime, columns: next regime, one-second steps), stationary distribution π and mean recurrence times $1/\pi_i$ (s). C: close following, P: stable platoon, F: free flow.

Dataset	From	→C	→P	→F	π_i	$1/\pi_i$
highD	Close following	0.970	0.028	0.002	0.454	2.20
	Stable platoon	0.028	0.961	0.011	0.435	2.30
	Free flow	0.013	0.039	0.948	0.111	9.03
NGSIM	Close following	0.746	0.245	0.009	0.096	10.42
	Stable platoon	0.032	0.920	0.048	0.730	1.37
	Free flow	0.004	0.202	0.794	0.174	5.75



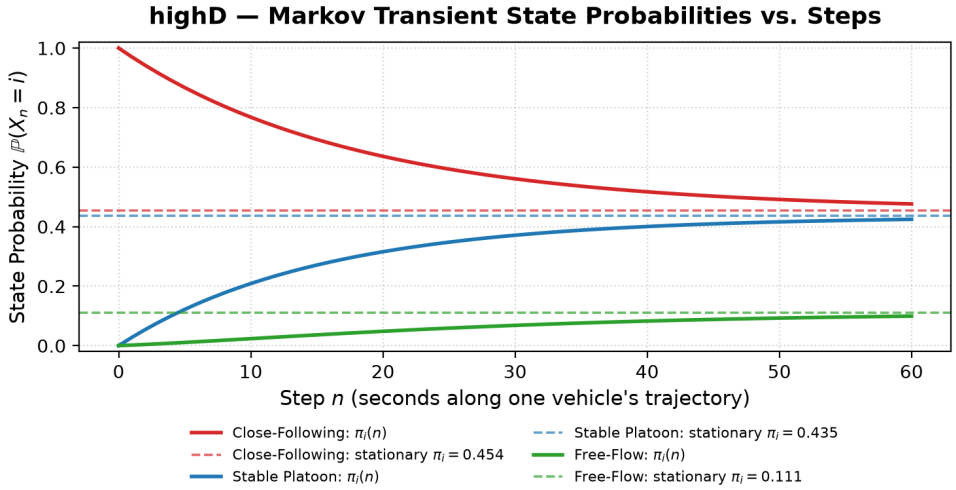
(a) highD **(b)** NGSIM

Figure 4: State-transition diagrams of the regime Markov chains. Arrow labels are transition probabilities p_{ij} per second; "stay" is p_{ii}.

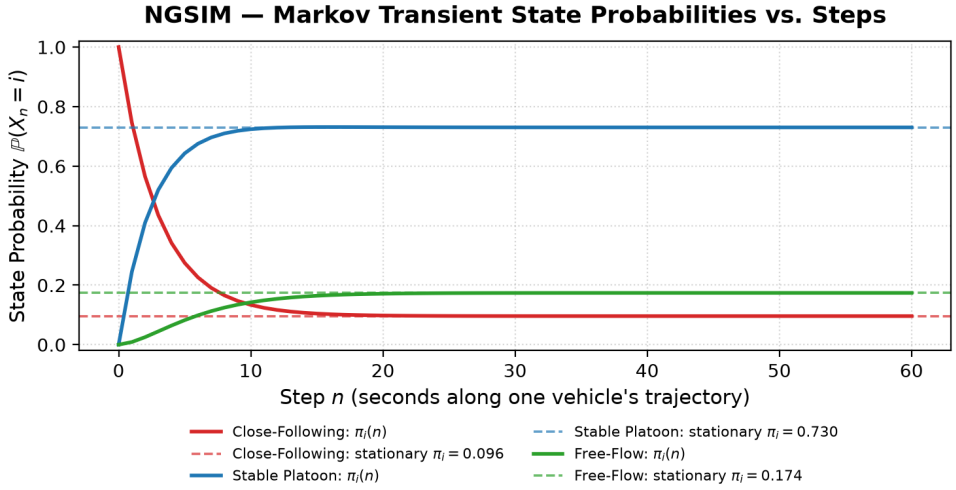
7.2 Real data results

7.2.1 Descriptive statistics

The headway samples are summarised in Table 1. highD headways are short and concentrated (mean 2.21 s, median 1.66 s, standard deviation 1.73 s); NGSIM headways are longer and more dispersed (mean 3.08 s, median 2.56 s, standard deviation 2.08 s). Both distributions are right-skewed, with the mean above the median.



(a) highD



(b) NGSIM

Figure 5: Transient regime probabilities $\pi(n) = \pi(0)P^n$ from close following (n in seconds), with the stationary distributions (dashed).

7.2.2 Posterior parameter estimates

Tables 6 and 7 list the posterior means for the seven distributions under the corrected (truncated) likelihood, under the paper's likelihood, and as published in [4]. For the proposed distribution the highD mode is $a = 0.954$ s (95% credible interval [0.950, 0.959]; published 0.936) with $b = 0.540$ ([0.539, 0.540]; published 0.540), i.e. a decay rate $\kappa = 0.617$ s^{-1}; on NGSIM $a = 2.336$ s ([2.326, 2.345]; published 2.277), $b = 0.524$ ([0.522, 0.527]; published 0.481) and $\kappa = 0.646$ s^{-1} (Table 8). The highD estimates are close to the published ones, and the NGSIM estimates are less so, as expected from the different NGSIM sample (Section 7.3.6). The two likelihoods give the same estimates for the proposed distribution and (to the third decimal) for the shifted exponential rate λ on both datasets, but different estimates for the other baselines on highD: for example the Weibull shape is 1.020 under the corrected likelihood and 1.450 under the paper likelihood, and the corrected gamma shape 1.117 is close to the exponential value 1. The 95% credible intervals are narrow because the multinomial likelihood treats the samples as independent; they understate the real uncertainty because successive headways of a follower are strongly dependent (Section 7.3.3).

Table 6: highD: posterior-mean parameters under the corrected (truncated) likelihood, the paper's likelihood, and as published in [4]. Parameter names are those of Table 2. Under the truncated likelihood the shifted-exponential threshold γ is not identified (its posterior is its prior).

Distribution	Parameter	Corrected	Paper likelihood	Published
Proposed	a	0.954	0.954	0.936
	b	0.540	0.540	0.540
Shifted Log-normal	μ	0.462	0.179	0.233
	σ	0.755	0.921	0.899
	γ	−0.001	0.438	0.377
Weibull	α	1.020	1.450	1.481
	β	1.744	2.472	2.473
Log-logistic	α	2.217	2.618	2.574
	β	1.565	1.717	1.719
Gamma	α	1.117	2.304	2.335
	β	0.628	1.038	1.055
Burr	α	1.464	3.777	3.199
	β	3.012	0.468	0.602
	λ	3.645	1.168	1.296
Shifted Exponential	λ	0.585	0.585	0.584
	γ	*not identified*	0.500	0.500

Table 7: NGSIM (I-80, first period): posterior-mean parameters, as in Table 6. The corrected shifted log-normal threshold γ lies at its upper bound $\alpha = 0.5$ s; published values of γ above 0.5 are not attainable with our smallest observed headway of 0.5 s.

Distribution	Parameter	Corrected	Paper likelihood	Published
Proposed	a	2.336	2.336	2.277
	b	0.524	0.524	0.481
Shifted Log-normal	μ	0.748	0.748	0.683
	σ	0.614	0.614	0.594
	γ	0.500	0.500	0.528
Weibull	α	1.533	1.678	1.744
	β	3.269	3.485	3.305
Log-logistic	α	3.618	3.664	3.910
	β	2.613	2.617	2.515
Gamma	α	3.479	3.693	4.175
	β	1.135	1.196	1.428
Burr	α	4.916	4.896	5.237
	β	0.517	0.533	0.524
	λ	2.066	2.086	2.021
Shifted Exponential	λ	0.387	0.387	0.423
	γ	*not identified*	0.500	0.558

Table 8: Posterior summary of the proposed distribution (corrected likelihood): mean, standard deviation, 95% equal-tailed credible interval, split $\hat{R}$ and effective sample size (ESS) from 2 chains $\times$ 5 000 draws.

Dataset	Parameter	Mean	SD	95% interval	$\hat{R}$	ESS
highD	a	0.9545	0.0021	[0.9504, 0.9586]	1.001	1,231
	b	0.5398	0.0003	[0.5391, 0.5405]	1.001	1,361
NGSIM	a	2.3356	0.0049	[2.3259, 2.3448]	1.004	1,249
	b	0.5241	0.0012	[0.5218, 0.5266]	1.001	1,343

7.2.3 Goodness of fit

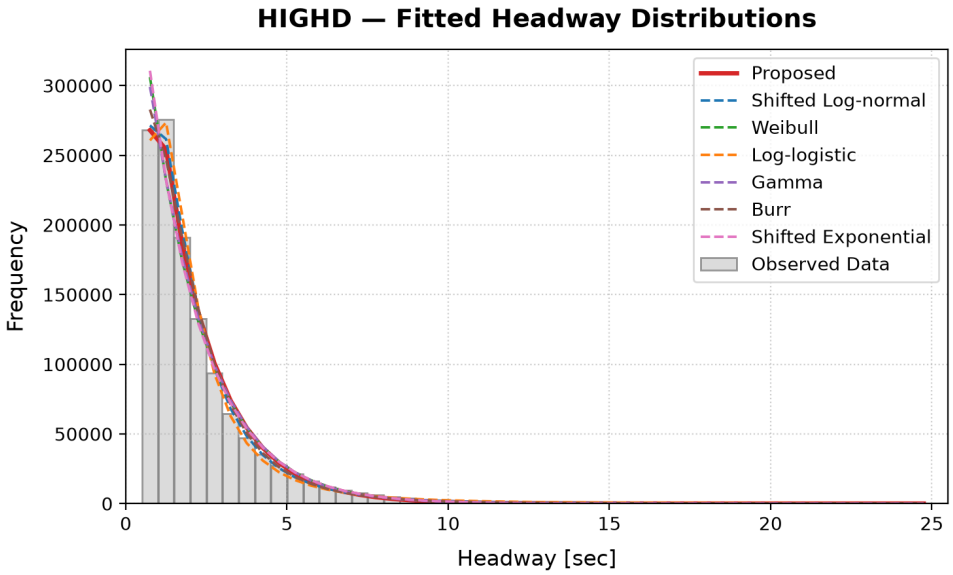
Figure 6 overlays the fitted densities (corrected likelihood) on the observed histograms. Table 9 gives the Kullback–Leibler divergence, Table 10 the distance W_{vec} of Eq. (4.12), and Table 11 the conventional distance W_{time} of Eq. (4.13) in seconds, each for the corrected fits, the fits with the paper's likelihood and the published values (where the study reports them).

highD. With the paper's likelihood the proposed distribution has the smallest KL divergence (0.0036) and the smallest W_{vec}, as published; the shifted log-normal (0.0063) and the shifted exponential (0.0080) follow. With the corrected likelihood the shifted log-normal is best by all three measures (KL 0.0026, W_{time} 0.048 s) and the proposed distribution is second by KL (0.0036), third by W_{vec} (after the shifted log-normal and the Burr distribution) and fourth by W_{time} (0.072 s). The baseline distributions improve markedly: the KL of the Weibull falls from 0.0586 to 0.0079, of the gamma from 0.0498 to 0.0075, of the Burr from 0.0288 to 0.0043 and of the log-logistic from 0.0229 to 0.0092, while the proposed distribution does not change. The conclusion that the proposed distribution is the best fit on highD therefore holds under the published likelihood and does not hold once the likelihood is conditioned on the observation window (Section 7.3.2). The proposed distribution remains competitive: its divergence is of the same order as that of the best distribution (0.0036 against 0.0026) and below those of the other five distributions, and it has two parameters where the best has three.

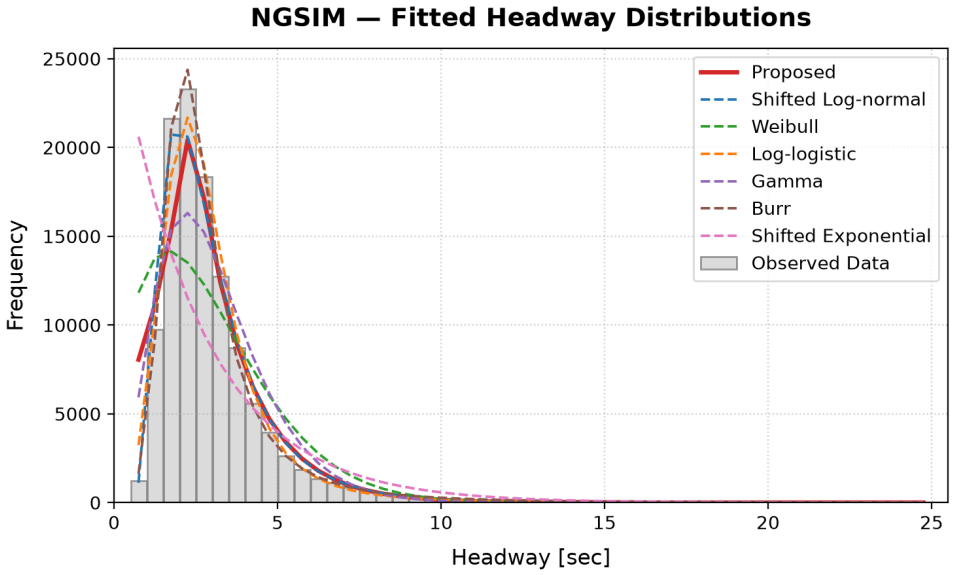
NGSIM. The Burr distribution is best by all three measures under both likelihoods (KL

0.0015, W_{vec} 0.0008, W_{time} 0.031 s), the shifted log-normal and log-logistic follow in KL (0.0153 and 0.0167), and the proposed distribution is fourth by all three (KL 0.0760, W_{time} 0.225 s). The worst fit is the shifted exponential by KL (0.2713) and W_{time} (0.645 s), and the Weibull by W_{vec} (0.0083). The likelihood hardly matters on NGSIM, because the headways are longer and the baselines put little mass below 0.5 s: the largest KL change is from 0.1944 to 0.1871 (Weibull).

Figure 6b suggests why the proposed distribution loses on NGSIM and Section 7.2.4 returns to it. The two Wasserstein variants agree on the winner under the corrected likelihood (the shifted log-normal on highD, the Burr distribution on NGSIM), but they do not agree on lower ranks (Table 10 against Table 11); for example on highD the log-logistic is fourth by W_{vec} (0.0013) but last by W_{time} (0.130 s), which illustrates that the sorted-probability form is blind to where a mismatch lies.



(a) highD



(b) NGSIM

Figure 6: Observed headway histogram (bars; frequency per 0.5 s bin) with the seven fitted distributions (corrected likelihood); curves show the expected count per bin of the window-conditioned model.

Table 9: Kullback–Leibler divergence D_{KL} (unitless, lower is better; best of each of our columns in bold). "Corrected": window-conditioned likelihood; "Paper": the likelihood of [4]; "Published": Table 5 of [4].

	highD			NGSIM		
Distribution	Corrected	Paper	Published	Corrected	Paper	Published
Proposed	0.0036	**0.0036**	0.0032	0.0760	0.0760	0.0805
Shifted Log-normal	**0.0026**	0.0063	0.0111	0.0153	0.0153	0.0157
Weibull	0.0079	0.0586	0.0534	0.1871	0.1944	0.2098
Log-logistic	0.0092	0.0229	0.0287	0.0167	0.0168	0.0173
Gamma	0.0075	0.0498	0.0454	0.1142	0.1150	0.1182
Burr	0.0043	0.0288	0.0332	**0.0015**	**0.0016**	0.0027
Shifted Exponential	0.0080	0.0080	0.0065	0.2713	0.2713	0.3024

Table 10: Wasserstein distance W_{vec} of sorted bin-probability vectors, Eq. (4.12) (unitless; lower is better; best of each of our columns in bold), with the same columns as Table 9. This is the reading of Eq. 17 of [4] whose scale matches the published values.

	highD			NGSIM		
Distribution	Corrected	Paper	Published	Corrected	Paper	Published
Proposed	0.0012	**0.0012**	0.0011	0.0043	0.0043	0.0040
Shifted Log-normal	**0.0008**	0.0015	0.0016	0.0022	0.0022	0.0021
Weibull	0.0020	0.0058	0.0052	0.0083	0.0090	0.0088
Log-logistic	0.0013	0.0022	0.0026	0.0021	0.0020	0.0020
Gamma	0.0019	0.0050	0.0049	0.0064	0.0063	0.0173
Burr	0.0011	0.0033	0.0032	**0.0008**	**0.0009**	0.0011
Shifted Exponential	0.0021	0.0021	0.0018	0.0050	0.0050	0.0047

Table 11: Conventional one-dimensional Wasserstein distance W_{time} on the time axis, Eq. (4.13), in seconds (lower is better; best of each column in bold), for the corrected and the paper-likelihood fits. The published study does not report this quantity.

	highD		NGSIM	
Distribution	Corrected	Paper	Corrected	Paper
Proposed	0.072	0.072	0.225	0.225
Shifted Log-normal	**0.048**	0.100	0.155	0.154
Weibull	0.072	0.347	0.461	0.506
Log-logistic	0.130	0.166	0.156	0.165
Gamma	0.080	0.290	0.351	0.359
Burr	0.059	0.228	**0.031**	**0.031**
Shifted Exponential	0.064	**0.064**	0.645	0.644

The statistical tests in Table 12 reject every distribution at the 5% level on both datasets in the chi-square test. In the two-sample KS test the corrected Burr distribution on NGSIM has the smallest statistic ($D = 0.011$) but its p-value ($\approx 7 \times 10^{-4}$) is also below 0.05. With samples of 10^5 to 10^6 observations even small departures are statistically significant, so the tests do not discriminate between the models, and the KS statistic D, the KL divergence and the Wasserstein distances are more informative. The published study reports the same chi-square result (all distributions rejected) but KS p-values close to 1 for most distributions, whereas ours are essentially zero; the sample sizes used for its KS test are not stated, and that part of its table

is not reproduced. Our KS p-values treat the 1 Hz samples as independent, which they are not (Section 7.3.3), so they are, if anything, too small.

Table 12: Goodness-of-fit tests of the corrected fits. KS D_1: one-sample statistic against the fitted distribution; KS D_2 and p_2: two-sample statistic against a simulated sample of $\min(n, 50\,000)$ draws and its p-value; p_χ: chi-square p-value. A value below 10^{-300} is printed as such.

	highD				NGSIM			
Distribution	D_1	D_2	p_2	p_χ	D_1	D_2	p_2	p_χ
Proposed	0.025	0.028	1.4×10^{-32}	$<10^{-300}$	0.076	0.076	3.7×10^{-175}	$<10^{-300}$
Shifted Log-normal	0.013	0.016	8.0×10^{-11}	$<10^{-300}$	0.030	0.032	1.7×10^{-31}	$<10^{-300}$
Weibull	0.041	0.042	3.0×10^{-73}	$<10^{-300}$	0.127	0.127	$<10^{-300}$	$<10^{-300}$
Log-logistic	0.016	0.018	3.2×10^{-13}	$<10^{-300}$	0.026	0.027	2.6×10^{-22}	$<10^{-300}$
Gamma	0.037	0.037	5.6×10^{-57}	$<10^{-300}$	0.086	0.089	1.3×10^{-242}	$<10^{-300}$
Burr	0.024	0.023	6.0×10^{-23}	$<10^{-300}$	0.010	0.011	6.8×10^{-4}	2.5×10^{-49}
Shifted Exponential	0.044	0.045	1.4×10^{-83}	$<10^{-300}$	0.228	0.228	$<10^{-300}$	$<10^{-300}$

7.2.4 highD against NGSIM

The proposed distribution is better suited to highD than to NGSIM, and the figure suggests why. Its density is $b^{|t-a|}$, which falls at the same rate κ on both sides of the mode. On highD the mode is at 0.954 s, only about 0.45 s above the truncation point $\alpha = 0.5$ s, so that the left branch is short and the symmetric form costs little. On NGSIM the mode is at 2.336 s and the left branch extends over about 1.8 s; the observed histogram rises more quickly to the mode than it decays after it, and a symmetric density with $\kappa = 0.646$ s^{-1} overestimates the shortest headways (Figure 6b). The three-parameter Burr, shifted log-normal and log-logistic distributions can represent this asymmetry; the Burr shape parameter on NGSIM is $\alpha = 4.916$, a peaked density with a heavy right tail. This explanation is read from the fitted curves and the Wasserstein and KL values and has not been tested by an asymmetric variant of the proposed density. In terms of traffic, the highD distribution is essentially a decay from a short mode near 0.95 s, whereas the NGSIM distribution has a distinct mode near 2.3 s and a steep rise to it, with fewer very short headways than a symmetric law implies.

7.3 Model validation

Table 13 summarises what was verified, at which level (Section 6.11), and what the result was. The subsections below give the details that are not in earlier tables.

Table 13: Summary of verification and validation by level.

Level	Check	Result
Mathematical	Closed-form CDF against numerical integration; all seven densities integrate to one; CDF derivative equals density	Pass (automated tests)
Simulation	MCMC recovers the known parameters of the proposed distribution from a sample drawn from it; truncated likelihood recovers a truncated sample's parameters better than the paper's	Pass (automated tests)
Simulation	Simulated renewal moments against Eq. (4.5)	Mean within 0.2%, variance within 7.0%; Table 4
Simulation	Simulated Markov chain reproduces its own π and P	Pass: π within TV 0.014 (highD), 0.008 (NGSIM) of the data (Table 18)
Empirical	Fit of seven distributions to highD and NGSIM	Tables 9–12; no model adequate by hypothesis tests
Empirical	Agreement with the published results	Paper-likelihood KL within 0.0058 (highD) and 0.0311 (NGSIM); identical NGSIM ranking; highD ranking depends on the likelihood
Empirical	Convergence of the 28 posteriors	All split $\hat{R} \leq 1.006$ (Table 14)
Empirical	Independence of successive one-second headways	Rejected (lag-1 0.906 and 0.727; Table 15)
Empirical	Independence of detector inter-arrival times	Weak dependence (lag-1 0.031 highD, 0.125 NGSIM; Table 15)
Empirical	Count dispersion at a virtual detector against renewal theory and Poisson	Observed 0.96 against renewal 0.77 and Poisson 1 on highD (Table 17)
Empirical	Markov property: Chapman–Kolmogorov and second-order test	Adequate for highD at short lags; not for NGSIM (Table 18)

7.3.1 MCMC convergence

Table 14 gives the largest split $\hat{R}$ over the parameters of each fit under both likelihoods, the smallest effective sample size and the acceptance rate of the corrected run; Figure 7 and Figure 8 show the posterior densities and the traces of the proposed distribution. All 28 fits (7 distributions $\times$ 2 datasets $\times$ 2 likelihoods) have $\hat{R} \leq 1.006$, the smallest ESS in the corrected runs is 437 (NGSIM) and 609 (highD) for chains of 10 000 retained draws, and the acceptance rates are between 19.1% and 26.5%, close to the optimum of about 23% [30].

The earlier non-convergence of the shifted exponential on highD. An earlier run of the paper likelihood with a random walk directly in (λ, γ) gave $\hat{R} = 4.92$ for the highD shifted exponential, and the other 13 fits converged. The shifted exponential has a feature that none of the other six has. For the bins above γ, the model probability of bin k is $p_k = e^{\lambda\gamma} e^{-\lambda e_k}(1 - e^{-\lambda w})$ with e_k the lower bin edge and $w = 0.5$ s, so that the binned log-likelihood contains the term $N\lambda\gamma$ and is *linear* in γ, increasing without bound until γ reaches its upper limit, the smallest observation $t_{\min}$ (the prior is $U(-10, t_{\min})$). The posterior of γ is then close to an exponential density pressed against that limit with a scale of roughly $1/(N\lambda)$, of the order of 10^{-6} s for the highD sample, while the posterior spread of λ is of the order of $\lambda/\sqrt{N} \approx 5 \times 10^{-4}$ and λ is coupled to γ through the same likelihood. A random walk in (λ, γ) with a covariance learned during warmup is poorly matched to a ridge this thin and this close to a boundary. The sampler now works in $v = \log(t_{\min} - \gamma)$, in which the posterior is no longer pressed against a boundary,

with the exact Jacobian e^v. Re-running the highD shifted exponential with the paper likelihood and this reparametrisation gives $\hat{R} = 1.005$ (the largest of all highD fits under that likelihood), posterior mean $\lambda = 0.585$ and $\gamma = 0.500$ s, which is at $t_{\min}$ as the diagnosis predicts. A regression test with a highD-sized sample checks that $\hat{R} < 1.05$ and that $\gamma < t_{\min}$ for every draw. The $\hat{R} = 4.92$ figure belongs to the earlier sampler and is not a result of this paper; the diagnosis was inferred from the likelihood and is supported by the success of the reparametrisation, not proved by it.

Table 14: MCMC convergence of the real-data fits: largest split $\hat{R}$ over the parameters under the corrected ("Trunc.") and paper ("Paper") likelihood, smallest effective sample size (ESS) and mean acceptance rate of the corrected run. Two chains with 5 000 post-warmup draws each.

	highD				NGSIM			
Distribution	$\hat{R}$ Trunc.	$\hat{R}$ Paper	ESS	Acc.	$\hat{R}$ Trunc.	$\hat{R}$ Paper	ESS	Acc.
Proposed	1.001	1.001	1,231	25.6%	1.004	1.004	1,249	26.5%
Shifted Log-normal	1.002	1.003	951	21.0%	1.005	1.003	437	19.1%
Weibull	1.003	1.001	1,242	25.0%	1.002	1.002	1,181	25.0%
Log-logistic	1.004	1.002	1,286	25.2%	1.003	1.001	1,233	25.2%
Gamma	1.001	1.001	1,066	24.3%	1.003	1.002	1,282	25.7%
Burr	1.001	1.000	845	22.4%	1.002	1.006	936	22.5%
Shifted Exponential	1.002	1.005	609	22.3%	1.004	1.004	539	21.4%

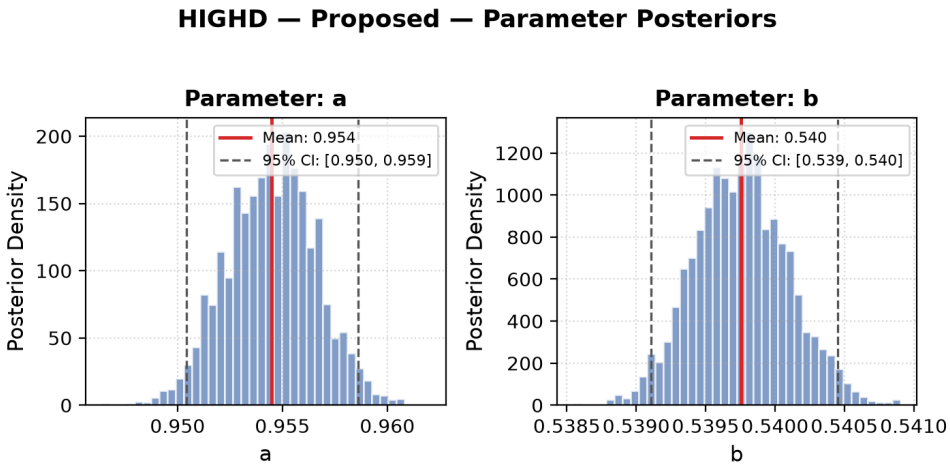
7.3.2 Truncation and the likelihood

The headways are retained only in $[0.5, 25]$ s, but the published likelihood treats each distribution on its natural support: the probabilities of the 49 bins sum to less than one, which is equivalent to assuming that the probability mass outside the window is a part of the data that happens to be unobserved, and penalises any distribution that puts mass there. For a sample that was truncated to the window, the exact likelihood is the conditioned one, Eq. (4.1). The two differ when the distribution puts appreciable mass outside the window, for example below 0.5 s, and not at all when it puts none there. The proposed distribution is defined on $[\alpha, \infty)$ and puts none below α, so the paper's likelihood favours it relative to the baselines, which are defined on $(0, \infty)$ or (γ, ∞). The paper's fits also pair a likelihood that is not exact for the sample with metrics that evaluate the window-conditioned model, so that fitting and evaluation use different distributions.

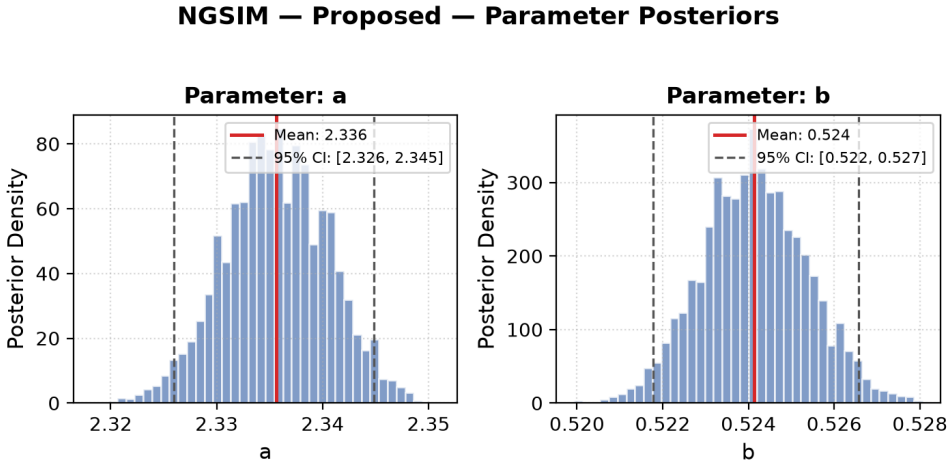
Re-estimating all fourteen fits with the conditioned likelihood quantifies the effect. On NGSIM it is small (the largest change in KL is for the Weibull, from 0.1944 to 0.1871). On highD it is large: the KL divergences of the Weibull, gamma, Burr, log-logistic and shifted log-normal fall by factors of 2.4 to 7.4 (Table 9; the shifted exponential, whose rate is unaffected, does not change), and the distributions are reordered: the proposed distribution moves from first to 2nd place by KL. The corrected posterior means of the baselines differ from the published ones (Table 6) for the same reason. The paper's method is therefore reproduced under the paper's likelihood (the purpose of Section 7.3.6), while the corrected likelihood is the one to be used for inference.

7.3.3 Independence of successive headways

The renewal process of Section 5.3 assumes independent headways. The autocorrelation of the sequence of one-second headways of a follower, computed only between samples of the same trajectory run, is given in Table 15. On highD it is 0.906 at lag 1 and 0.601 at lag 5 (95% white-noise band ± 0.0019), on NGSIM 0.727 and 0.426. Successive one-second headways of a

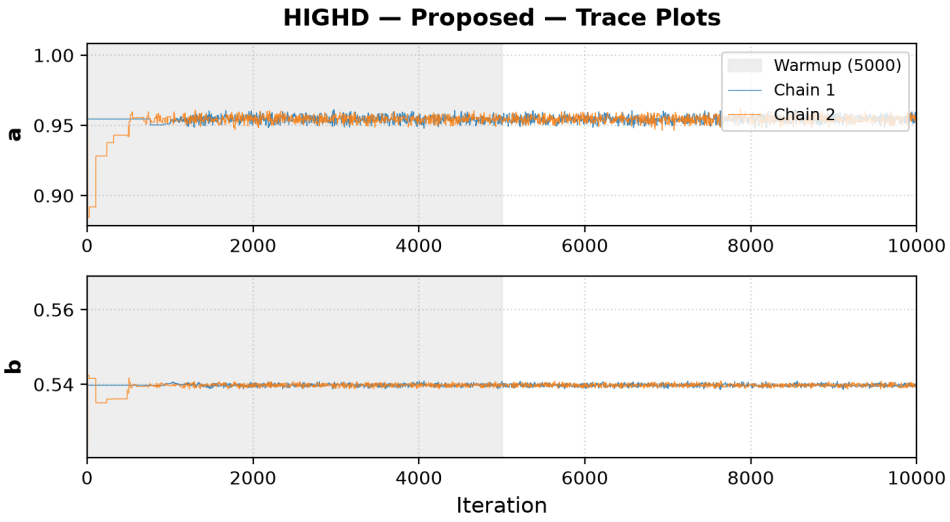


(a) highD

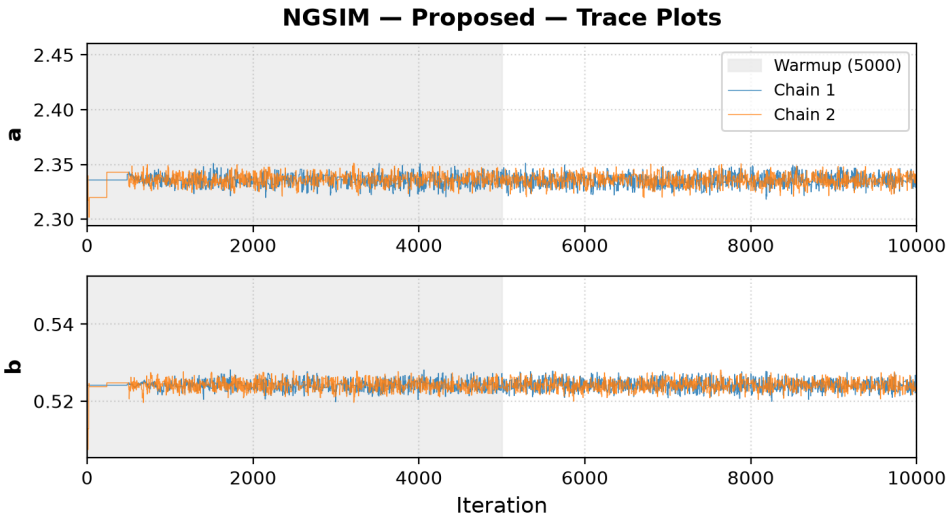


(b) NGSIM

Figure 7: Posterior distributions of the parameters a (s) and b of the proposed distribution (corrected likelihood).



(a) highD



(b) NGSIM

Figure 8: Trace plots of the two chains for the proposed distribution over all 10 000 iterations; the shaded part is warmup.

follower are therefore strongly dependent and the dependence decays over several seconds, as expected since a driver's gap changes gradually. This is a property of the headway *sample*: it shows that the 1 Hz sample cannot be treated as an i.i.d. sample of one vehicle's behaviour, and that the renewal simulation of Section 7.1, which resamples headways independently, models the headway *distribution* and how counts accumulate under independence, and not the temporal structure of the data. It does not by itself show that detector inter-arrival times are dependent; Section 7.3.4 examines those directly.

Table 15: Autocorrelation at lags 1–5 of the one-second time headways within trajectory runs ("Headways") and of the detector inter-arrival times of Section 7.3.4 ("Detector gaps"; lag in vehicles), with the half-width of the 95% white-noise band.

	Headways (1 Hz)		Detector gaps	
Lag	highD	NGSIM	highD	NGSIM
1	0.906	0.727	0.031	0.125
2	0.820	0.614	−0.009	0.091
3	0.740	0.535	0.017	0.091
4	0.667	0.473	0.000	0.013
5	0.601	0.426	0.008	0.034
95% band ±	0.0019	0.0058	0.0060	0.0434

7.3.4 Detector inter-arrival times: testing the renewal model

Detector crossing times were reconstructed from the trajectories as described in Section 6.10. Table 16 compares them with the headway-driven model. On highD there are 108,842 inter-arrival times from 337 lane-recordings, with mean 3.08 s and a flow of 1,168 veh/h/lane; the headway-driven renewal rate was 1,627 veh/h/lane. On NGSIM there are 2,046 inter-arrival times from 6 lanes during the first period, mean 2.64 s, 1,364 veh/h/lane against the headway-driven 1,167. The headway-driven rates are therefore not detector flows: they are *lower* than the detector flow on NGSIM and *higher* on highD. They are reciprocals of the mean headway of followers that have a leader, sampled once per second per follower, and not of the time between vehicles passing a point, so they need not agree; the difference was not decomposed further.

Table 16: Detector inter-arrival times reconstructed from the trajectories (all gaps, no window applied). Std: standard deviation; CV^2: squared coefficient of variation of the pooled gaps (pooling lanes and recordings with different flows inflates it); Flow: 3600/mean gap; the last two columns are the shares of gaps below 0.5 s and above 25 s.

Dataset	Gaps	Mean (s)	Median (s)	Std (s)	CV^2	Flow (veh/h/lane)	< 0.5 s	> 25 s
highD	108,842	3.08	2.11	3.34	1.18	1,168	0.4%	0.2%
NGSIM	2,046	2.64	2.30	1.44	0.30	1,364	0.3%	0.0%

Independence. The detector gaps are almost uncorrelated on highD: lag-1 autocorrelation 0.031 (white-noise band ±0.0060) and about zero at later lags (Table 15, Figure 9b); the value at lag 1 lies outside the band, but it is small, and the independence assumption of the renewal model is a reasonable approximation for the *detector* process on highD, in contrast with the 1 Hz headways. On NGSIM the lag-1 value is 0.125 (band ±0.0434) and 0.091 at lag 2, a modest positive dependence that is consistent with platooning. The sample is small (2,046 gaps, one period at one detector position).

Count dispersion. Table 17 compares the variance-to-mean ratio of counts in consecutive

windows with the renewal prediction from the same series' gaps and with Poisson. On highD in one-minute windows (5,383 windows from 334 series, mean count 19.5) the observed ratio is 0.96, against 0.77 predicted by the renewal process built on the detector gaps and 1 for Poisson. The observed counts are thus much closer to Poisson than to the headway-driven prediction (0.61), and closer to Poisson than to the renewal process of the same gaps. The excess over the renewal value is consistent with flow that drifts during a recording, which is not part of the renewal model, so the observed ratio probably overstates the stationary dispersion; even so, the regularity predicted by the headway-driven model (variance-to-mean 0.61) is not seen in highD counts. On NGSIM the observed ratio is 0.52 at 60 s, well below Poisson, as the headway-driven model also says (0.46), against 0.29 for the renewal process of the detector gaps, from 86 windows only. The agreement on NGSIM is close for the headway-driven value but it should be read with caution: it is a single short period and the detector gaps themselves have a different CV^2 (0.30, Table 16) from the headway sample (0.46).

Table 17: Index of dispersion (variance/mean) of vehicle counts in consecutive windows of w seconds at the virtual detector: observed, renewal prediction σ_u^2/μ_u^2 from the same series' gaps (series-wise, window-weighted average), and Poisson. Mean count: average vehicles per window per lane.

Dataset	w (s)	Windows	Mean count	Observed	Renewal	Poisson
highD	30	10,986	9.8	0.94	0.77	1
	60	5,383	19.5	0.96	0.77	1
	120	2,493	39.5	0.97	0.76	1
NGSIM	30	176	11.3	0.41	0.29	1
	60	86	22.6	0.52	0.29	1
	120	42	45.1	0.64	0.29	1

Summary. The detector reconstruction supports two statements and contradicts one. It supports an approximately independent arrival process on highD and a regular (sub-Poisson) one on NGSIM; it contradicts the idea that the headway-driven renewal model, with the dispersion of the headway distribution, describes the variability of arrival counts: that dispersion (0.61 on highD) is not the dispersion of detector gaps nor of counts. The limitations are one detector position, strongly non-stationary short recordings, and (for NGSIM) a single 15-minute period.

7.3.5 Markov chain

Table 18 gives the validation of the Markov chain of Section 5.4 on the within-run regime sequences. All quantities use only pairs and triples of samples of the same run. The stationary distribution of the estimated P lies within total-variation distance 0.014 (highD) and 0.008 (NGSIM) of the observed regime fractions; the small difference is consistent with within-run transition counting, which omits the last sample of each run from the row totals. A long simulated path of the chain reproduces its stationary distribution and re-estimates P to within 0.0013 (highD) and 0.0064 (NGSIM), which is a simulation verification of the code. *Chapman–Kolmogorov.* For a first-order chain the empirical n-step transition matrix equals P^n (Eq. (4.7)). The largest absolute error is small on highD at $n = 2$ (0.006) and grows to 0.055 at $n = 5$; on NGSIM it is 0.092 at $n = 2$ and 0.226 at $n = 5$, which is large for probabilities. *First against second order.* The likelihood-ratio test rejects the first-order chain on both datasets ($G^2 = 1046.4$ and 4291.5 with 12 degrees of freedom), as is unavoidable with this many transitions, so the effect size is the useful quantity: the mean total-variation distance between $P(\cdot \mid i, j)$ and $P(\cdot \mid j)$ is 0.0014 on highD, a negligible departure, and 0.0400 on NGSIM, a small but real one. The first-order chain is therefore an adequate description of the regime sequence on highD for short horizons and a poorer one on NGSIM, where the Chapman–Kolmogorov errors are large and the history

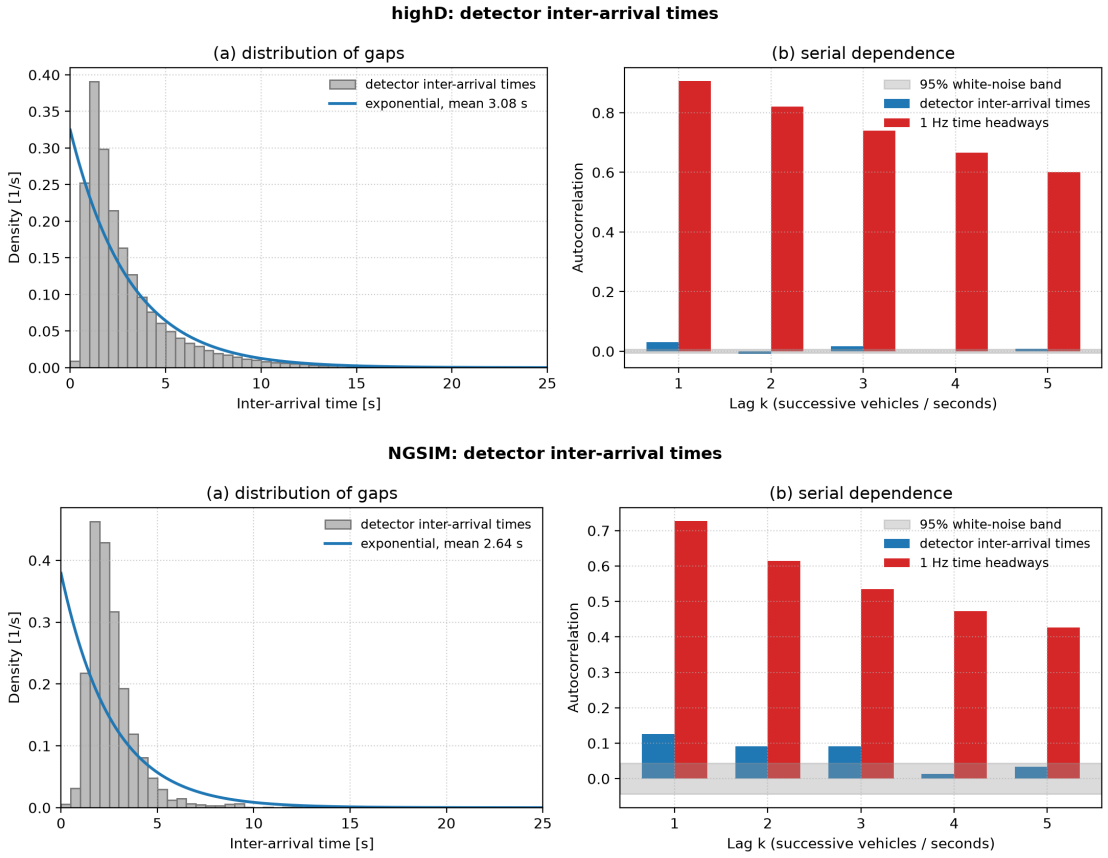


Figure 9: Detector inter-arrival times. (a) density of the gaps (s) with the exponential law of the same mean; (b) autocorrelation of the gaps at lags 1–5 (vehicles) and of the 1 Hz time headways within trajectory runs (seconds), with the 95% white-noise band of the gaps.

matters (a semi-Markov or second-order model would be the natural extension).

Table 18: Validation of the first-order regime Markov chain on within-run sequences. TV: total-variation distance; G^2: likelihood-ratio statistic of first against second order.

Quantity	highD	NGSIM
Pairs / transitions used	1,112,619	113,861
Chapman–Kolmogorov max error, $n = 2$	0.006	0.092
Chapman–Kolmogorov max error, $n = 3$	0.018	0.158
Chapman–Kolmogorov max error, $n = 5$	0.055	0.226
Stationary π vs observed regime fractions (TV)	0.0144	0.0084
Simulated chain: occupation vs π (TV)	0.0123	0.0039
Simulated chain: max error of re-estimated P	0.0013	0.0064
Second-order test G^2 (12 d.f.)	1046.4	4291.5
Second-order test p-value	2.0×10^{-216}	$<10^{-300}$
Second-order effect size (mean TV)	0.0014	0.0400

7.3.6 Comparison with the published results

With the paper's likelihood, the KL divergences differ from the published Table 5 by at most 0.0058 on highD (Log-logistic) and 0.0311 on NGSIM (Shifted Exponential), and W_{vec} by at most 0.0006 on highD (Weibull) and 0.0110 on NGSIM (Gamma; the other six differ by 0.0003 or less). The published W_{vec} for gamma on NGSIM (0.0173) is inconsistent with its own KL ranking, where gamma has a lower divergence than Weibull but twice the distance, and we do not expect any consistent implementation to reproduce it. The ranking by KL on NGSIM, from Burr to the shifted exponential, is identical to the published ranking. On highD our ranking with the paper's likelihood differs from the published one only in that the shifted exponential and the shifted log-normal swap second and third place (published: shifted exponential 0.0065, shifted log-normal 0.0111; ours 0.0080 and 0.0063). The remaining differences are largest for the badly fitting baselines (the shifted exponential and the Weibull on NGSIM), whose divergence is sensitive to the exact sample. The published NGSIM sample could not be reproduced: it differs from ours in source (reconstructed against original trajectories), and probably in the selection of lanes and times, which the paper does not state. The published and reproduced parameters agree closely on highD for the proposed distribution ($a = 0.954$ against 0.936, b identical) and the shifted exponential, and less so for the shifted log-normal and Burr, whose parameters are weakly identified under the paper likelihood. On NGSIM $b = 0.524$ against the published 0.481 is the largest difference for the proposed distribution. The qualitative conclusions of [4] for these two datasets are reproduced under the paper's likelihood: the proposed distribution is best on highD and the Burr distribution is best on NGSIM. The first is lost under the conditioned likelihood (Section 7.3.2). The published KS p-values are not reproduced.

7.4 Overall discussion and traffic interpretation

Traffic regimes. The two datasets are in different states. The highD sample is dominated by short headways: the stationary distribution puts 0.454 on close following and 0.435 on stable platoon, and the regimes are persistent (the model-implied expected sojourn in close following is 33.7 s). Followers at one-to-four-second headways are the normal state of the highway lane. NGSIM I-80 is dominated by platoon-range headways (0.730) with few very short ones (0.096), and its regimes are less persistent (close following lasts 3.9 s in expectation); the typical NGSIM follower keeps a headway of two to three seconds. The flows implied by the mean headway (1,627 and 1,167 veh/h/lane) are reciprocals of the mean headway of followers that have a leader; they

describe car-following states and not flows observed at a detector (1,168 and 1,364 veh/h/lane).

Arrivals: regular, or nearly Poisson? The headway-driven renewal model, with the headway distribution as inter-arrival law, implies that counts are far less variable than Poisson (variance-to-mean ratios 0.61 and 0.46; a Poisson model would overestimate the variability by a factor of about 1.6 to 2.2). This is what car following would be expected to produce, since a follower adjusts its gap and so suppresses both very short and very long headways. The detector data qualify it. On NGSIM the counts are indeed sub-Poisson (0.52 at 60 s). On highD the one-minute counts are close to Poisson (0.96), presumably because the gaps between successive vehicles at a point vary strongly with the lane and the moment (CV^2 of pooled gaps 1.18, and 0.77 on average within a lane-recording). For applications that use arrival counts, such as queueing and signal-timing analysis, the practical message is that a Poisson model is a reasonable description of highD-type uninterrupted flow at the minute scale, whereas in the I-80 sample it overstates the variability, and that the regularity should be taken from detector gaps, not from followers' headways.

Which distribution? No distribution is best everywhere: the Burr distribution wins on NGSIM, the shifted log-normal wins on highD under the conditioned likelihood and the proposed distribution wins on highD under the paper's likelihood. The practical lesson is the dependence of the ranking on the dataset and on the fitting method, and not the identity of the single best model. The proposed distribution has the advantage of two interpretable parameters (the mode and the decay rate), a closed-form distribution function that supports exact sampling, and a posterior that is easy to sample, at the cost of a symmetry constraint that matters when the mode is far from the lower threshold.

What the results do not show. Because the goodness-of-fit tests reject every model, the results show relative and not absolute adequacy. The agreement of simulated renewal counts with renewal theory verifies the simulation, not the independence assumption, which the 1 Hz data contradict (and the detector data only weakly on highD). The uncertainty reported by the posterior is conditional on independent samples and is optimistic.

8 Limitations and Future Work

8.1 Limitations of the present study

1. **Likelihood.** All uncertainty statements use a binned multinomial likelihood that treats the 1 Hz samples as independent, which they are not (lag-1 autocorrelation 0.906 on highD); credible intervals and KS p-values are therefore too narrow. The binning also discards within-bin information, exact only to the 0.5 s bin width.
2. **Published reproduction.** The NGSIM sample of the published study is not reproduced; the first recording period of the Data.gov export was chosen from the published parameters, which is an inference. The published KS p-values and the published Wasserstein value for the gamma distribution on NGSIM are not reproduced.
3. **Wasserstein distance.** Eq. 17 of the published study is ambiguous. W_{vec} matches its scale and is blind to the position of the bins; W_{time} is the conventional distance. The two rank models differently; both are reported.
4. **Goodness-of-fit tests.** All distributions are rejected by the chi-square test on samples of 10^5–10^6 observations; the tests do not select a model.

5. **Renewal model.** The headway-driven process uses time headways between a follower and its leader and not detector inter-arrival times; its i.i.d. resampling removes the serial dependence of the data. The detector validation uses one virtual detector per recording (highD) or one position and one 15-minute period (NGSIM: 2,046 gaps from 6 lanes), and flow drifts within recordings.
6. **Regimes and Markov chain.** The thresholds (1.5 s and 4 s) are modelling choices; sojourn times are censored by the observation window; the first-order assumption is adequate on highD but not on NGSIM.
7. **Single run.** The MCMC and the simulations were run with one seed.
8. **Scope.** Two human-driven datasets, pooled over lanes, vehicle types and speeds, with no covariates. The mixed human–automated datasets of the original study (Waymo, Lyft) and exiD are not included.

8.2 Future work

These are proposals, not implemented work. (i) Replace the binned independent likelihood by a model that accounts for serial dependence (for example a hidden Markov or semi-Markov model with the regimes as states), and check how much the posterior widens. (ii) Give the proposed density different decay rates on its two sides, which the NGSIM fit suggests, and test whether this restores its competitiveness on NGSIM. (iii) Fit the seven distributions to detector inter-arrival times instead of 1 Hz headways, at several detector positions, and relate their parameters to flow. (iv) Make the headway distribution depend on speed and vehicle type, and add the mixed human–automated datasets of the original study. (v) Compare against a mixture or Cowan-type composite model, which the literature suggests for the combination of free and tracking headways.

9 Conclusion

Headways on two highway datasets were modelled with seven distributions fitted by MCMC, a renewal arrival process and a three-regime Markov chain. With the published likelihood the proposed distribution $f(t \mid a, b) \propto b^{|t-a|}$ is the best of the seven on highD by KL divergence (0.0036) and fourth on NGSIM (0.0760), where the Burr distribution is best (0.0015); these are the qualitative findings of the published study, and our divergences agree with its values to within 0.0058 and 0.0311. Conditioning the likelihood on the $[0.5, 25]$ s observation window, which the published method does not do, changes the highD result: the shifted log-normal is best (0.0026) and the proposed distribution second (0.0036), so the published advantage on highD comes in part from a likelihood that penalises distributions with mass below 0.5 s. A symmetry of the proposed density about its mode is a plausible reason for its weaker performance on NGSIM. All posteriors converge, and the non-convergence of the highD shifted exponential seen in an earlier run is removed by a reparametrisation of its threshold.

Simulated arrivals from the headways are markedly more regular than a Poisson stream (variance-to-mean ratios 0.61 and 0.46), and the simulated counts agree with renewal theory, but this verifies the simulation and not the model. Tested against detector counts reconstructed from the trajectories, the headway-driven model's regularity is not found on highD, where one-minute counts have a variance-to-mean ratio of 0.96, nearly Poisson, while on NGSIM the counts are sub-Poisson (0.52). The one-second headways of a follower are strongly autocorrelated (lag-1 0.906 on highD), but the detector inter-arrival times are nearly independent on highD. The stationary regime distributions are 0.454/0.435/0.111 (close, platoon, free) on highD and 0.096/0.730/0.174 on NGSIM; the first-order Markov chain describes the highD regime sequence well at short horizons and the NGSIM sequence less well. The main practical conclusions are

that rankings of headway distributions depend on the likelihood used for fitting, that a headway distribution does not by itself determine the variability of arrival counts, and that the validation of a stochastic model has to be carried out against data and not against the model's own theory.

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A Source Code

The listings below are the exact source of the functions in the project repository (`headway/` package); they are extracted automatically, not retyped.

```
def _proposed_pdf(t, a: float, b: float):
    """Normalized probability density function for g(t) = b^|t-a| on [ALPHA, inf).

    Paper Eq. (1), (12), (13):
        int_ALPHA^inf b^|t-a| dt = (b^(a-ALPHA) - 2) / ln(b)    if a > ALPHA
                                   -b^(ALPHA-a) / ln(b)          if a <= ALPHA
    """
    if b <= 0.0 or b >= 1.0:
        return np.zeros_like(t, dtype=float)
    t = np.asarray(t, dtype=float)
    out = np.zeros_like(t)
    m = t >= ALPHA
    if not np.any(m):
        return out
    log_b = np.log(b)
    if a > ALPHA:
        norm = (b ** (a - ALPHA) - 2.0) / log_b
    else:
        norm = -(b ** (ALPHA - a)) / log_b
    out[m] = (b ** np.abs(t[m] - a)) / norm
    return out
```

Listing 1: Density of the proposed headway distribution, normalised on $[\alpha, \infty)$ (`distributions.py`).

```
def _proposed_cdf(t, a: float, b: float):
    """Piecewise closed-form CDF on [ALPHA, inf), paper Eq. (2) and Eq. (9)-(13)."""
    if b <= 0.0 or b >= 1.0:
        return np.zeros_like(t, dtype=float)
    t = np.asarray(t, dtype=float)
    out = np.zeros_like(t)
    m = t >= ALPHA
    if not np.any(m):
        return out

    tm = t[m]
    res = np.zeros_like(tm)
    if a > ALPHA:
        denom = b ** (a - ALPHA) - 2.0
        m1 = tm <= a
        res[m1] = (b ** (a - ALPHA) - b ** (a - tm[m1])) / denom
        m2 = ~m1
        res[m2] = (b ** (a - ALPHA) + b ** (tm[m2] - a) - 2.0) / denom
    else:
        res = 1.0 - b ** (tm - ALPHA)

    out[m] = np.clip(res, 0.0, 1.0)
    return out
```

Listing 2: Distribution function of the proposed distribution, Eq. (5.3) (`distributions.py`).

```
def make_logpost(dist: Dist, counts: np.ndarray, tmin: float, likelihood: str = "paper
    ") -> Callable[[Sequence[float]], float]:
    """Construct the binned multinomial log-posterior function.

    ln p(theta | counts) = ln p(theta) + sum_{k=1}^49 O_k ln(p_k(theta))
    where p_k(theta) is the model probability in bin k.

    likelihood="paper" uses the distribution's own (un-truncated) density, as the
        paper does, so
    the bin probabilities sum to less than one. likelihood="truncated" conditions on
        the observed
    window [0.5, 25] s (they sum to one), which is the exact likelihood for a sample
        that was
    truncated to that window. Under it the shifted exponential's threshold is not
        identifiable
```

```
    (the conditional law does not depend on it) and its posterior is just the prior.
    """
    if likelihood not in LIKELIHOODS:
        raise ValueError(f"likelihood must be one of {LIKELIHOODS}")
    bin_probs = dist.raw_bin_probs if likelihood == "paper" else dist.bin_probs

    def logpost(theta: Sequence[float]) -> float:
        lp = dist.logprior(theta, tmin)
        if not np.isfinite(lp):
            return -np.inf
        if not dist.check_params(theta):
            return -np.inf
        with np.errstate(all="ignore"):
            p = bin_probs(theta)
        if not np.all(np.isfinite(p)) or np.any(p < 0.0) or p.sum() <= 0.0:
            return -np.inf
        # Multinomial log-likelihood over the 49 bins
        safe_p = np.maximum(p, 1e-300)
        return lp + float(np.dot(counts, np.log(safe_p)))

    return logpost
```

Listing 3: Binned log-posterior used by the sampler; `likelihood` selects the paper's or the truncated likelihood (`fit.py`).

```
def metropolis(
    logp: Callable[[np.ndarray], float],
    x0: Sequence[float],
    n_iter: int = 10000,
    n_warmup: int = 5000,
    rng: np.random.Generator = None,
) -> ChainOutput:
    """Run one MCMC chain using adaptive random-walk Metropolis-Hastings.

    Args:
        logp: Log-posterior function theta -> log p(theta | data).
        x0: Initial parameter vector.
        n_iter: Total iterations (warmup + retained).
        n_warmup: Number of warmup/burn-in iterations.
        rng: NumPy random Generator.

    Returns:
        ChainOutput with draws, warmup draws, full history, and acceptance rates.
    """
    if rng is None:
        rng = np.random.default_rng()

    x = np.array(x0, dtype=float)
    d = len(x)
    lp = logp(x)

    # If initial logp is non-finite, attempt small random perturbations to find valid
        support
    if not np.isfinite(lp):
        for _ in range(50):
            cand = x + 0.05 * rng.standard_normal(d)
            cand_lp = logp(cand)
            if np.isfinite(cand_lp):
                x, lp = cand, cand_lp
                break
        if not np.isfinite(lp):
            raise ValueError(f"Starting point {x0} has non-finite log-posterior {lp}")

    # Initial proposal covariance scaled by Gelman-Roberts-Gilks factor
    cov = np.diag((0.1 * np.maximum(np.abs(x), 0.5)) ** 2)
    L = np.linalg.cholesky(cov)
    scale = 2.38 / np.sqrt(d)

    full_draws = np.empty((n_iter, d), dtype=float)
    acc_window = 0
    acc_warmup_total = 0
    acc_post_total = 0
```

```
47
48      window_size = max(50, min(100, n_warmup // 4)) if n_warmup >= 100 else max(10,
            n_warmup // 2)
49      cov_interval = max(100, min(500, n_warmup // 2)) if n_warmup >= 200 else max(20,
            n_warmup)
50
51      for i in range(n_iter):
52          step = scale * (L @ rng.standard_normal(d))
53          y = x + step
54          lq = logp(y)
55
56          # Metropolis acceptance probability alpha = min(1, exp(lq - lp))
57          if np.isfinite(lq) and np.log(rng.random()) < (lq - lp):
58              x, lp = y, lq
59              acc_window += 1
60              if i < n_warmup:
61                  acc_warmup_total += 1
62              else:
63                  acc_post_total += 1
64
65          full_draws[i] = x
66
67          # Adaptation during warmup only (preserves ergodicity post-warmup)
68          if i < n_warmup:
69              if (i + 1) % window_size == 0:
70                  rate = acc_window / float(window_size)
71                  scale *= float(np.clip(np.exp((rate - 0.234) * 0.8), 0.5, 2.0))
72                  acc_window = 0
73
74              # Re-estimate proposal covariance structure from empirical warmup
                    trajectory
75              if (i + 1) % cov_interval == 0 and i >= window_size:
76                  history = full_draws[max(0, i - 2 * cov_interval) : i + 1]
77                  if np.all(history.std(axis=0) > 1e-8):
78                      emp_cov = np.cov(history.T)
79                      reg_cov = np.atleast_2d(emp_cov) + 1e-8 * np.eye(d)
80                      try:
81                          L = np.linalg.cholesky(reg_cov)
82                      except np.linalg.LinAlgError:
83                          pass
84
85      n_post = max(1, n_iter - n_warmup)
86      n_warm = max(1, n_warmup)
87      post_acc_rate = float(acc_post_total / n_post)
88      warm_acc_rate = float(acc_warmup_total / n_warm)
89
90      return ChainOutput(
91          draws=full_draws[n_warmup:],
92          warmup=full_draws[:n_warmup],
93          full_draws=full_draws,
94          acceptance_rate=post_acc_rate,
95          warmup_acceptance=warm_acc_rate,
96      )
```

Listing 4: Adaptive random-walk Metropolis–Hastings sampler (`mcmc.py`).

```
1   def _chain(task: Tuple[str, np.ndarray, float, np.ndarray, int, int, int, int, str])
        -> ChainOutput:
2       """Worker task executing one independent MCMC chain."""
3       name, counts, tmin, x0, chain_idx, n_iter, n_warmup, seed, likelihood = task
4       dist = DISTS[name]
5       logpost = make_logpost(dist, counts, tmin, likelihood)
6       rng = np.random.default_rng([seed, chain_idx, 100])
7
8       # Chain 0 starts at MAP; Chain 1+ starts slightly perturbed to verify convergence
9       start = np.array(x0, dtype=float)
10      if chain_idx > 0:
11          perturb = 1.0 + 0.05 * (1.0 if chain_idx % 2 == 1 else -1.0)
12          perturbed_start = start * perturb
13          if np.isfinite(logpost(perturbed_start)):
14              start = perturbed_start
15
16      i = GAP_PARAM.get(name)
```

```
    if i is None:
        return metropolis(logpost, start, n_iter=n_iter, n_warmup=n_warmup, rng=rng)

    # Sample v = log(min t - gamma); log p(v) = log p(gamma) + v is the exact change of variables.
    logpost_z = lambda z: logpost(_from_z(z, i, tmin)) + z[i]
    out = metropolis(logpost_z, _to_z(start, i, tmin), n_iter=n_iter, n_warmup=n_warmup, rng=rng)
    return replace(out, draws=_from_z(out.draws, i, tmin), warmup=_from_z(out.warmup, i, tmin),
                   full_draws=_from_z(out.full_draws, i, tmin))
```

Listing 5: One MCMC chain, including the log-gap coordinates for the shifted exponential (`fit.py`).

```
def segment_ids(vehicle: np.ndarray, second: np.ndarray) -> np.ndarray:
    """Label each retained 1 Hz sample with the run of consecutive seconds of one vehicle it belongs to.

    A new run starts whenever the vehicle changes or a second is missing (the follower had no
    valid leader, or its headway left [0.5, 25] s, in between). Markov transitions and
    autocorrelations are computed only between samples of the same run, so no transition spans
    two vehicles or a gap in time.
    """
    v, s = np.asarray(vehicle), np.asarray(second)
    new = np.ones(len(v), dtype=bool)
    new[1:] = (v[1:] != v[:-1]) | (s[1:] != s[:-1] + 1)
    return np.cumsum(new) - 1
```

Listing 6: Trajectory runs: consecutive seconds of one vehicle (`datasets/common.py`).

```
    @classmethod
    def from_headways(cls, headways: np.ndarray, groups: Optional[np.ndarray] = None) -> "TrafficRegimeMarkovChain":
        """Construct Markov Chain from consecutive empirical or simulated headway observations.

        `groups` are trajectory-run ids (datasets.common.segment_ids). Given them, a transition is
        counted only between two samples of the same run, i.e. consecutive seconds of one vehicle.
        Without them every adjacent pair counts, which is right only for a single trajectory.
        """
        h = np.asarray(headways, dtype=float)
        # Discretize into 3 physical car-following regimes
        # 0: Close-Following [0.5, 1.5), 1: Stable Platoon [1.5, 4.0), 2: Free-Flow [4.0, inf)
        states = np.digitize(h, [1.5, 4.0])  # returns 0, 1, 2
        states = np.clip(states, 0, 2)

        # Count 1-step transitions
        n_states = 3
        ok = np.ones(len(states) - 1, dtype=bool) if groups is None else (np.asarray(groups)[:-1] == np.asarray(groups)[1:])
        counts = np.bincount((states[:-1] * 3 + states[1:])[ok], minlength=9).reshape(3, 3).astype(float)

        # Regularize zero rows with uniform transitions
        row_sums = counts.sum(axis=1, keepdims=True)
        P = np.where(row_sums > 0, counts / row_sums, np.eye(n_states))
        return cls(P)
```

Listing 7: Markov transition estimation from within-run pairs only (`stochastic_process.py`).

```
def simulate_counts(headways: np.ndarray, times: Sequence[float], n_paths: int = 2000,
                    seed: int = 0) -> np.ndarray:
    """N(t) for n_paths independent renewal paths: headways bootstrapped i.i.d. from the data.
```

```
4
5       N(t) = max{n : S_n <= t} with S_n = H_1 + ... + H_n. Returns (n_paths, len(times))
            .
6       """
7       h = np.asarray(headways, float)
8       t_max = max(times)
9       k = int(np.ceil(2 * t_max / h.mean())) + 50  # about twice the arrivals needed,
            plus a buffer
10      arrivals = np.cumsum(np.random.default_rng(seed).choice(h, size=(n_paths, k)),
            axis=1)
11      if (arrivals[:, -1] <= t_max).any():
12          raise RuntimeError("not enough simulated arrivals to cover t_max")
13      return np.stack([(arrivals <= t).sum(axis=1) for t in times], axis=1)
```

Listing 8: Simulation of the renewal counting process $N(t)$ (`validation.py`).

```
1   def renewal_validation(headways: np.ndarray, times: Sequence[float] = (30, 60, 120,
        300),
2                          n_paths: int = 2000, seed: int = 0) -> Tuple[pd.DataFrame, np.
                               ndarray]:
3       """Simulated moments and distribution of N(t) against renewal theory.
4
5       With mu = E[H], sigma^2 = Var(H) the standard results are
6           E[N(t)]   ~ t/mu + (sigma^2 - mu^2) / (2 mu^2)
7           Var[N(t)] ~ sigma^2 t / mu^3,           N(t) approximately Normal.
8       A Poisson process of the same rate has Var = E; the ratio Var/E tends to sigma^2/
            mu^2.
9       ks_normal / ks_poisson are max |F_sim - F_model| for the count distribution.
10      """
11      h = np.asarray(headways, float)
12      mu, s2 = h.mean(), h.var(ddof=1)
13      counts = simulate_counts(h, times, n_paths, seed)
14      rows = []
15      for j, t in enumerate(times):
16          c = counts[:, j]
17          m_th, v_th = t / mu + (s2 - mu ** 2) / (2 * mu ** 2), s2 * t / mu ** 3
18          k = np.arange(c.min(), c.max() + 1)
19          f_sim = np.array([(c <= x).mean() for x in k])
20          rows.append({
21              "t_s": t,
22              "mean_sim": c.mean(), "mean_theory": m_th, "mean_rel_err": abs(c.mean() -
                    m_th) / m_th,
23              "var_sim": c.var(ddof=1), "var_theory": v_th, "var_rel_err": abs(c.var(
                    ddof=1) - v_th) / v_th,
24              "var_poisson": t / mu,
25              "fano_sim": c.var(ddof=1) / c.mean(), "fano_theory": s2 / mu ** 2,
26              "ks_normal": float(np.abs(f_sim - norm.cdf((k + 0.5 - m_th) / np.sqrt(v_th
                    ))).max()),
27              "ks_poisson": float(np.abs(f_sim - poisson.cdf(k, t / mu)).max()),
28          })
29      return pd.DataFrame(rows), counts
```

Listing 9: Simulated moments of $N(t)$ against renewal theory (`validation.py`).

```
1   def markov_validation(headways: np.ndarray, n_sim: int = 200_000, seed: int = 0,
        groups=None) -> Dict[str, float]:
2       """Does a first-order, time-homogeneous 3-state chain describe the regime sequence
            ?
3
4       * chapman_kolmogorov_n: max |empirical n-step matrix - P^n|   (exact for a Markov
            chain)
5       * stationary_*: stationary pi against the observed regime fractions
6       * sim_*: a simulated long path against pi, and P re-estimated from it against P
7       * second_order_*: likelihood-ratio test of first order against second order, plus
            its effect
8         size (average total-variation distance between P(.|i,j) and P(.|j)). With very
              large n the
9         test rejects for tiny effects, so read the effect size too.
10      """
11      s = regime_states(headways)
12      P = _row_normalise(_transition_counts(s, 1, groups))
```

```
    out: Dict[str, float] = {"n_transitions": float(_transition_counts(s, 1, groups).sum())}

    for n in (2, 3, 5):
        diff = np.abs(_row_normalise(_transition_counts(s, n, groups)) - np.linalg.matrix_power(P, n))
        out[f"chapman_kolmogorov_{n}_max_err"] = float(diff.max())

    A = np.vstack([P.T - np.eye(3), np.ones(3)])
    pi = np.clip(np.linalg.lstsq(A, np.append(np.zeros(3), 1.0), rcond=None)[0], 0, 1)
    pi /= pi.sum()
    frac = np.bincount(s, minlength=3) / len(s)
    out["stationary_max_abs_diff"] = float(np.abs(pi - frac).max())
    out["stationary_tv_distance"] = float(0.5 * np.abs(pi - frac).sum())

    rng = np.random.default_rng(seed)
    cum, u = np.cumsum(P, axis=1), rng.random(n_sim)
    sim = np.empty(n_sim, dtype=int)
    sim[0] = 0
    for i in range(1, n_sim):
        sim[i] = min(int(np.searchsorted(cum[sim[i - 1]], u[i], side="right")), 2)
    out["sim_occupation_tv_distance"] = float(0.5 * np.abs(np.bincount(sim, minlength=3) / n_sim - pi).sum())
    out["sim_P_reestimate_max_err"] = float(np.abs(_row_normalise(_transition_counts(sim)) - P).max())

    ok3 = _same_run(groups, len(s), 2)  # (i, i+2) in one run implies i+1 is too: runs are consecutive seconds
    n3 = np.bincount((s[:-2] * 9 + s[1:-1] * 3 + s[2:])[ok3], minlength=27).reshape(3, 3, 3).astype(float)
    n_ij, n_jk, n_j = n3.sum(2), n3.sum(0), n3.sum((0, 2))
    with np.errstate(divide="ignore", invalid="ignore"):
        g2 = 2 * np.nansum(np.where(n3 > 0, n3 * np.log(n3 * n_j[None, :, None]
                                                        / (n_ij[:, :, None] * n_jk[None, :, :])), 0.0))
        cond_ij = n3 / n_ij[:, :, None]
        cond_j = n_jk / n_j[:, None]
        tv = 0.5 * np.nansum(np.abs(cond_ij - cond_j[None, :, :]), axis=2)
    out["second_order_g2"] = float(g2)
    out["second_order_dof"] = 12.0
    out["second_order_p"] = float(chi2.sf(g2, 12))
    out["second_order_effect_tv"] = float((np.nan_to_num(tv) * n_ij).sum() / n_ij.sum())
    return out
```

Listing 10: Chapman–Kolmogorov, stationarity and first- against second-order checks (`validation.py`).

```
def crossing_times(vehicle: np.ndarray, t: np.ndarray, pos: np.ndarray, lane: np.ndarray, x0: float):
    """Time at which each vehicle first crosses the cross-section ``pos == x0``.

    Rows must be sorted by (vehicle, t). A crossing is a pair of consecutive samples of the same
    vehicle with ``pos`` on opposite sides of ``x0``; the crossing time is interpolated linearly
    between them and the lane is the lane of the later sample. A vehicle is counted once, at its
    first crossing. Returns (vehicle id, crossing time, lane) arrays.
    """
    same = vehicle[1:] == vehicle[:-1]
    a, b = pos[:-1] - x0, pos[1:] - x0
    cross = same & (((a < 0) & (b >= 0)) | ((a > 0) & (b <= 0)))
    i = np.flatnonzero(cross)
    frac = (-a[i]) / (b[i] - a[i])
    tc = t[i] + frac * (t[i + 1] - t[i])
    v, ln = vehicle[i + 1], lane[i + 1]
    first = np.ones(len(i), dtype=bool)
    first[1:] = v[1:] != v[:-1]  # i is increasing, so a vehicle's crossings are adjacent
    return v[first], tc[first], ln[first]
```

Listing 11: Virtual-detector crossing times by linear interpolation, Eq. (6.1) (`detector.py`).

```python
def window_count_summary(gaps: pd.DataFrame, window: float = 60.0, min_windows: int = 5) -> Dict[str, float]:
    """Observed variability of vehicle counts in consecutive windows against renewal and Poisson theory.

    For every (unit, lane) the arrivals are counted in consecutive windows of ``window`` seconds, from the
    first arrival to the last. The index of dispersion (variance/mean) of the counts is compared with
    the renewal prediction sigma^2/mu^2 computed from the *same series'* inter-arrival times, and with 1
    (Poisson). Series are averaged with weights equal to their number of windows. Flow that drifts during a
    recording inflates the observed dispersion, which is why the result is a bound and not an exact test.
    """
    rows = []
    for (unit, lane), s in gaps.groupby(["unit", "lane"]):
        t = s["t"].to_numpy()
        t = np.concatenate([[t[0] - s["gap"].iloc[0]], t])  # first arrival, recovered from the first gap
        n_win = int((t[-1] - t[0]) // window)
        if n_win < min_windows or len(s) < 10:
            continue
        c = np.histogram(t, bins=t[0] + window * np.arange(n_win + 1))[0]
        gp = s["gap"].to_numpy()
        if c.mean() > 0:
            rows.append((n_win, c.mean(), c.var(ddof=1), gp.var(ddof=1) / gp.mean() ** 2, window / gp.mean()))
    r = np.array(rows)
    w = r[:, 0]
    fano_obs = float(np.average(r[:, 2] / r[:, 1], weights=w))
    return {
        "window_s": window, "n_series": len(r), "n_windows": int(w.sum()),
        "mean_count": float(np.average(r[:, 1], weights=w)),
        "fano_observed": fano_obs,
        "fano_renewal_theory": float(np.average(r[:, 3], weights=w)),
        "fano_poisson": 1.0,
        "mean_count_renewal": float(np.average(r[:, 4], weights=w)),
    }
```

Listing 12: Count dispersion in fixed windows against renewal and Poisson theory (`detector.py`).